

Team Note of HCMUS - Stellar

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```

brute < input.inp > output.ans
fc output.out output.ans > log.txt || call :WA
)
echo Accepted!
pause
exit

:WA
echo Wrong Answer
echo Test case:
type input.inp
echo Participant answer:
type output.out
echo Jury answer:
type output.ans

pause
exit

```

1.3 Stress Tester (Ubuntu)

```

/*
    chmod +x run.sh
    ./run.sh
*/

#!/bin/bash
g++ -O3 -std=c++17 gen.cpp -o gen || exit 1
g++ -O3 -std=c++17 brute.cpp -o brute || exit 1
g++ -O3 -std=c++17 sol.cpp -o sol || exit 1

for ((i = 1; ; ++i)); do
    ./gen > input.txt
    ./brute < input.txt > brute_out.txt
    ./sol < input.txt > sol_out.txt

    if ! diff -w brute_out.txt sol_out.txt > /dev/null; then
        echo "=====
        echo "--> WA o test $i!"
        echo "Input:"
        cat input.txt
        echo "Expected Output (Brute):"
        cat brute_out.txt
        echo "Your Output (Sol):"
        cat sol_out.txt
        echo "=====
        break
    fi
    echo "Passed test $i"
done

```

1.4 Bench Marker

```

@echo off
title Bench Marker 6000
echo Compiling files, please wait...
g++ -O2 -Wl,--stack=268435456 -std=c++11 -o test.exe test.cpp
echo Done compiling test generator!
g++ -O2 -Wl,--stack=268435456 -std=c++11 -o sol.exe sol.cpp
echo Done compiling (probably) faulty submission!

for /l %x in (1, 1, 1000) do (
    echo Test case numero: %x
    test > input.inp
    sol < input.inp > output.out
)

pause
exit

```

1.5 Min Plus Convolution

```

// a is convex a[i+1]-a[i] <= a[i+2]-a[i+1]
vector<int> min_plus_convolution(vector<int>& a, vector<int>& b) {
    int n = sz(a), m = sz(b);
    vector<int> c(n + m - 1, INF);
    auto dc = [&](auto Y, int l, int r, int jl, int jr) {

```

```

        if (l > r) return;
        int mid = (l + r) / 2, from = -1, & best = c[mid];
        for (int j = jl; j <= jr; ++j)
            if ((int i = mid - j; i >= 0 && i < n)
                if (best > a[i] + b[j])
                    best = a[i] + b[j], from = j;
        Y(Y, l, mid - 1, jl, from);
        Y(Y, mid + 1, r, from, jr);
    };
    return dc(dc, 0, n - 1 + m - 1, 0, m - 1), c;
}

```

1.6 Mathematics

Equations

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, c \neq 1$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

1.7 CP Sublime 3 Build

```

{
    "encoding": "utf-8",
    "working_dir": "${file_path}",
    "shell_cmd": "g++ -Wall -O2 -std=c++14 -o ${file_base_name}.exe ${file_base_name}.cpp",
    "file_regex": "^(\\.([^\:]*):([0-9]+)?(?:[0-9]+)?(?:\\.*)$)",
    "selector": "source.c++,source.c",
    "variants":
    [
        {
            "name": "Run",
            "shell_cmd": "g++ -Wall -O2 -std=c++14 \"${file}\" -o \"${file_base_name}\" && start cmd /c \"\"${file_path}/${file_base_name}\" & pause\"
        }
    ]
}

```

1.8 CP Sublime 3 Keymap

```

[
    { "keys": ["f9"], "command": "build" },
    { "keys": ["ctrl+shift+x"], "command": "toggle_comment", "args": { "block": false } },

```

```
{ "keys": ["ctrl+shift+c"], "command": "toggle_comment",
  "args": { "block": false } },

{ "keys": ["ctrl+f12"], "command": "sort_lines", "args":
{"case_sensitive": false} },
{ "keys": ["ctrl+shift+f12"], "command": "sort_lines",
"args": {"case_sensitive": true} },
]
```

2 Data Structure

2.1 Line Container

```
struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};
```

2.2 Convex Hull Trick

```
/*
CHT template
*/

const ll INF = 1e18 + 69;

struct CHT{
    #define Node pair<ll, ll>
    vector<Node> a;
    vector<double> bubble;
    bool flag;

    // if flag = 0: if inserted in increasing order, return
    min, otherwise return max
    // flag = 1: the opposite

    CHT(bool _flag){
        flag = _flag;
    }

    double getInter(Node a, Node b){
        double x = (double) (b.second - a.second) / (a.first -
        b.first);
        return x;
    }

    void add(Node x, bool isMin = 1){
        if (a.empty()) {a.push_back(x); return;}
        if (a.back().first == x.first){
            if (isMin) minimize(a[a.size() - 1].second,
            x.second);
            else maximize(a[a.size() - 1].second, x.second);
            return;
        }
        while(a.size() >= 2){
```

```
double x1 = getInter(a.back(), x), x2 =
bubble.back();
if (flag){
    if (x1 >= x2) break;
}
else{
    if (x1 <= x2) break;
}
a.pop_back(); bubble.pop_back();
}
bubble.push_back(getInter(a.back(), x));
a.push_back(x);
}

ll get(ll x){
    if (a.empty()) return INF;
    long idx;
    if (flag) idx = lower_bound(ALL(bubble), x) -
bubble.begin();
    else idx = lower_bound(ALL(bubble), x,
greater<double>()) - bubble.begin();
    return a[idx].first * x + a[idx].second;
}
};
```

2.3 Fast Segment Tree

```
struct SegmentTree{
    int n;
    vector<int> a;

    SegmentTree(int _n){
        n = _n;
        a.resize(n * 2 + 2, -INF);
    }

    void update(int i, int v){
        i += n; if (!maximize(a[i], v)) return;
        while(i > 1){
            i >>= 1;
            a[i] = max(a[i * 2], a[i * 2 + 1]);
        }
    }

    int get(int l, int r){
        l += n; r += n + 1;
        int ans = -INF;
        while(l < r){
            if (l & 1) maximize(ans, a[l++]);
            if (r & 1) maximize(ans, a[--r]);
            l >>= 1; r >>= 1;
        }
        return ans;
    }
};
```

2.4 Lichao Tree

```
/*
Lichao Tree
*/

const ll INF = 1e18;

struct LichaoTree{
    #define Node pair<ll, ll>

    long n;
    bool isMin;
    vector<Node> a;

    LichaoTree(long _n, bool flag = 1){
        n = _n;
        isMin = flag;
        if (flag) a.resize(n * 4 + 4, make_pair(0, INF));
        else a.resize(n * 4 + 4, make_pair(0, -INF));
    }

    ll f(Node t, ll x){return t.first * x + t.second;}

    void update(Node x, long l, long r, long id){
```

```

    bool check1 = f(a[id], l) >= f(x, l), check2 =
    f(a[id], r) >= f(x, r);
    bool check3 = f(a[id], l) <= f(x, l), check4 =
    f(a[id], r) <= f(x, r);
    if (check1 && check2){
        if (isMin) a[id] = x;
        return;
    }
    if (check3 && check4){
        if (!isMin) a[id] = x;
        return;
    }
    if (l == r) return;
    long mid = (l + r) >> 1;
    update(x, l, mid, id * 2);
    update(x, mid + 1, r, id * 2 + 1);
}

void update(Node x){update(x, 1, n, 1);}

ll get(long i, long l, long r, long id){
    ll ans1 = f(a[id], i);
    if (l == r) return ans1;
    long mid = (l + r) >> 1;

    ll ans2;
    if (i <= mid) ans2 = get(i, l, mid, id * 2);
    else ans2 = get(i, mid + 1, r, id * 2 + 1);

    if (isMin) return min(ans1, ans2);
    else return max(ans1, ans2);
}

ll get(long i){return get(i, 1, n, 1);}
};

```

2.5 Persistent Segment Tree

```

class PersistentSegmentTree {
private:
    int n;
    vector<Node*> roots; // Store roots of different versions

    // Build initial segment tree
    Node* build(int start, int end) {
        if (start == end) return new Node(0); // Initialize
        with zero; customize as needed
        int mid = (start + end) / 2;
        return new Node(0, build(start, mid), build(mid + 1,
        end));
    }

    // Create new version with point update
    Node* update(Node* prev, int start, int end, int idx, int
    val) {
        if (start == end) return new Node(prev->value + val);
        // Update node value
        int mid = (start + end) / 2;
        if (idx <= mid)
            return new Node(prev->value + val,
            update(prev->left, start, mid, idx, val),
            prev->right);
        else
            return new Node(prev->value + val, prev->left,
            update(prev->right, mid + 1, end, idx, val));
    }

    // Query for a range [L, R]
    int query(Node* node, int start, int end, int L, int R) {
        if (start > R || end < L) return 0; // Return 0 for
        sum; adjust for other ops
        if (L <= start && end <= R) return node->value;
        int mid = (start + end) / 2;
        return query(node->left, start, mid, L, R) +
        query(node->right, mid + 1, end, L, R);
    }

public:
    PersistentSegmentTree(int size) : n(size) {
        roots.push_back(build(0, n - 1)); // Initial version
    }
}

```

```

// Update tree version by creating a new version
void update(int idx, int val) {
    roots.push_back(update(roots.back(), 0, n - 1, idx,
    val));
}

// Query a specific version
int query(int version, int L, int R) {
    return query(roots[version], 0, n - 1, L, R);
}
};

```

2.6 Treap

```

/*
Treap template
*/

struct BST{
    struct Node{
        int val, priority;
        Node* child[2];
        Node(int val): val(val), priority(rngesus(0, MASK(30)
        - 1)){
            child[0] = child[1] = nullptr;
        }
    };

    Node* root;

    BST(){
        root = nullptr;
    }

    void split(Node* cur, int x, Node* & L, Node*& R){ //
    split up to a point x
        if (cur == nullptr) {L = R = nullptr;return;}
        if (x < cur->val){
            split(cur->child[0], x, L, cur->child[0]);
            R = cur;
        }
        else{
            split(cur->child[1], x, cur->child[1], R);
            L = cur;
        }
    }

    void merge(Node*& cur, Node* L, Node* R){
        if (L == nullptr) cur = R;
        else if (R == nullptr) cur = L;
        else{
            if (R->priority > L->priority) {
                cur = R;
                merge(cur->child[0], L, R->child[0]);
            }
            else{
                cur = L;
                merge(cur->child[1], L->child[1], R);
            }
        }
    }

    void insert(int x){
        Node *L, *R;
        split(root, x, L, R);
        merge(root, L, new Node(x));
        merge(root, root, R);
    }

    void erase(int x){
        Node *L = nullptr, *mid = nullptr, *R = nullptr;
        split(root, x-1, L, mid);
        split(mid, x, mid, R);

        merge(root, L, R);
    }

    bool find(int x){
        Node *id = root;
        while(true){
            if (id == nullptr) return false;
            if (id->val == x) return true;
        }
    }
}

```

```

        else if (x < id -> val) id = id -> child[0];
        else id = id -> child[1];
    }
}

void clear(){
    while(root != nullptr){
        int val = root -> val;
        erase(val);
    }
}
};

```

3 Geometry

3.1 Point Struct

```

template<class T>
struct Point {
    T x, y;
    explicit Point(T x = 0, T y = 0) : x(x), y(y) {}

    bool operator<(Point p) const { return tie(x, y) <
        tie(p.x, p.y); }
    bool operator==(Point p) const { return tie(x, y) ==
        tie(p.x, p.y); }

    Point operator+(Point p) const { return Point(x + p.x, y +
        p.y); }
    Point operator-(Point p) const { return Point(x - p.x, y -
        p.y); }
    Point operator*(T d) const { return Point(x * d, y * d); }
    Point operator/(T d) const { return Point(x / d, y / d); }
    Point operator-() const { return Point(-x, -y); }

    T dot(Point p) const { return x * p.x + y * p.y; }
    T cross(Point p) const { return x * p.y - y * p.x; }
    T cross(Point a, Point b) const { return (a -
        *this).cross(b - *this); }

    T dist2() const { return x * x + y * y; }
    double dist() const { return sqrt((double)dist2()); }
    Point unit() const { return *this / dist(); }
    Point perp() const { return Point(-y, x); }
    double angle() const { return atan2(y, x); }

    friend istream& operator>>(istream& is, Point& p) { return
        is >> p.x >> p.y; }
    friend ostream& operator<<(ostream& os, const Point& p) {
        return os << p.x << " " << p.y; }
};

```

3.2 Boiler Plate

```

/*
    Just some good old boiler plate code
*/

struct PointRect{
    double x, y;
    PointRect(){}
    PointRect(double _x, double _y){
        x = _x, y = _y;
    }
};

struct PointPolar{
    double theta, len;
    PointPolar(){}
    PointPolar(double _theta, double _len){
        theta = _theta; len = _len;
    }
};

const double PI = acos(0) * 2;

PointPolar toPolar(PointRect p){
    double x = p.x, y = p.y;
    double len = sqrt(p.x * p.x + p.y * p.y);
    if (x== 0){

```

```

        if (y>= 0) return PointPolar(PI / 2, sqrt(p.x * p.x +
            p.y * p.y));
        return PointPolar(PI * 3 / 2, sqrt(p.x * p.x + p.y *
            p.y));
    }
    if (y == 0){
        if (x > 0) return PointPolar(0, sqrt(p.x * p.x + p.y *
            p.y));
        return PointPolar(PI, sqrt(p.x * p.x + p.y * p.y));
    }

    double more = 0;
    while(x < 0 || y < 0){
        more += PI / 2;
        swap(x, y);
        y = -y;
    }
    return PointPolar(more + atan(y / x), len);
}

PointRect toRect(PointPolar p){
    double theta = p.theta, len = p.len;
    int cnt = 0;
    while(theta >= PI / 2) {
        theta -= PI / 2; cnt++;
    }
    PointRect v = PointRect(cos(theta) * len, sin(theta) *
        len);
    while(cnt--){
        v.y = -v.y;
        swap(v.x, v.y);
    }
    return v;
}

double getArea(PointRect a, PointRect b, PointRect c){
    return a.x * (b.y - c.y) + b.x * (c.y - a.y) + c.x * (a.y
        - b.y);
}

array<double, 3> getFunction(pair<PointRect, PointRect> x){
    array<double, 3> ans;
    ans[0] = x.first.y - x.second.y;
    ans[1] = x.second.x - x.first.x;
    ans[2] = -(ans[0] * x.first.x + ans[1] * x.first.y);
    return ans;
}

PointRect getIntersection(pair<PointRect, PointRect> line1,
    pair<PointRect, PointRect> line2){
    array<double, 3> slope1 = getFunction(line1), slope2
        = getFunction(line2);
    double delta = (slope1[1] * slope2[0] - slope2[1] *
        slope1[0]);
    if (abs(delta) <= EPS) return PointRect(INF * 2, INF * 2);
    double y = -(slope1[2] * slope2[0] - slope2[2] *
        slope1[0]) / delta;
    double x = (slope1[2] * slope2[1] - slope2[2] * slope1[1])
        / delta;
    return PointRect(x, y);
}

void find_convex_hull(vector<PointRect> &a){
    if (a.size() <= 2) return;
    sort(ALL(a), [] (PointRect x, PointRect y){
        if (x.x != y.x) return x.x < y.x;
        return x.y < y.y;
    });

    PointRect l = a[0], r = a.back();
    vector<PointRect> up, down;
    up.push_back(l); down.push_back(l);
    for(long i = 1; i<a.size(); ++i){
        if (getArea(l, r, a[i]) >= 0){
            while(up.size() > 1 && getArea(up[up.size() - 2],
                up[up.size() - 1], a[i]) > 0) up.pop_back();
            up.push_back(a[i]);
        }
        if (getArea(l, r, a[i]) <= 0){

```

```

        while(down.size() > 1 && getArea(down[down.size()
        - 2], down[down.size() - 1], a[i]) < 0)
            down.pop_back();
        down.push_back(a[i]);
    }
}
a = up;
for(long i = down.size() - 2; i>=1; --i)
    a.push_back(down[i]);
}

```

3.3 Circles Intersect

```

/*
 * Usage :
 * Computes the pair of points at which two circles intersect.
 * -> out
 * Returns false in case of no intersection.
 */

typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>* out)
{
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) /
    d2);
    *out = {mid + per, mid - per};
    return true;
}

```

3.4 Circle vs CCW Polygon Intersection

```

/*
 * Usage :
 * Returns the area of the intersection of a circle with a CCW
 * polygon.
 * Time complexity : O(n)
 */

typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b =
        (p.dist2()-r*r)/d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
    };
    auto sum = 0.0;
    rep(i,0,sz(ps))
        sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
    return sum;
}

```

3.5 Circle vs Line Intersect

```

/*
 * Usage :
 * Finds the intersection between a circle and a line.
 * Returns a vector of either 0, 1, or 2 intersection points.
 * P is intended to be Point<double>.
 */

template<class Point>
vector<Point> circleLine(Point c, double r, Point a, Point b)
{
    Point ab = b - a, p = a + ab * (c-a).dot(ab) / ab.dist2();
    double s = a.cross(b, c), h2 = r*r - s*s / ab.dist2();
    if (h2 < 0) return {};
    if (h2 == 0) return {p};
    Point h = ab.unit() * sqrt(h2);

```

```

    return {p - h, p + h};
}

```

3.6 Complex Numbers Utils

Time Complexity:

Example

point a(3, 2), b(2, -7);

```

// vector addition and subtraction
cout << a + b << endl; // (5,-5)
cout << a - b << endl; // (1,9)

```

```

// scalar multiplication
cout << 3.0 * a << endl; // (9,6)
cout << a / 5.0 << endl; // (0.6,0.4)

```

Functions using std::complex

What else can we do with complex numbers? Well, there's a lot that is really easy to code.

Vector addition: a + b

Scalar multiplication: r * a

Dot product: (conj(a) * b).x

Cross product: (conj(a) * b).y

notice: conj(a) * b = (ax*bx + ay*by) + i (ax*by - ay*bx)

Squared distance: norm(a - b)

Euclidean distance: abs(a - b)

Angle of elevation: arg(b - a)

Slope of line (a, b): tan(arg(b - a))

Polar to cartesian: polar(r, theta)

Cartesian to polar: point(abs(p), arg(p))

Rotation about the origin: a * polar(1.0, theta)

Rotation about pivot p: (a-p) * polar(1.0, theta) + p

UPD: added more useful functions

Angle ABC: abs(remainder(arg(a-b) - arg(c-b), 2.0 * M_PI))

remainder normalizes the angle to be between [-PI, PI]. Thus, we can get the positive non-reflex angle by taking its abs value.

Project p onto vector v: v * dot(p, v) / norm(v);

Project p onto line (a, b): a + (b - a) * dot(p - a, b - a) / norm(b - a)

Reflect p across line (a, b): a + conj((p - a) / (b - a)) * (b - a)

Intersection of line (a, b) and (p, q):

```

point intersection(point a, point b, point p, point q) {
    double c1 = cross(p - a, b - a), c2 = cross(q - a, b - a);
    return (c1 * q - c2 * p) / (c1 - c2); // undefined if
    parallel
}

```

3.7 Hull Diameter

```

/*
 * Usage :
 * Returns the two points with max distance on a convex hull
 * (CCW, no duplicate/collinear points).
 * Time complexity : O(n)
 */

```

```

typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
    int n = sz(S), j = n < 2 ? 0 : 1;
    pair<ll, array<P, 2>> res({0, {S[0], S[0]}});

```



```

rep(i,0,j)
for (; j = (j + 1) % n) {
    res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]}});
    if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
        break;
}
return res.second;
}

```

3.8 Manhattan Minimum Spanning Tree

Time Complexity:

```

struct Path{
    int u, v; ll w;
    Path(){
        Path(int _u, int _v, ll _w){u = _u, v = _v, w = _w;}
};

typedef Point<ll> P;
const ll INF = 1e18;
namespace ManhattanSpanningTreeSolver {
    #define sz(v) ((int)v.size())

    vector<Path> solve(vector<P> ps) {
        vector<int> id(sz(ps));
        iota(ALL(id), 0);
        vector<Path> edges;
        for(int k = 0; k < 4; ++k) {
            sort(ALL(id), [&](int i, int j) {
                return (ps[i]-ps[j]).x < (ps[j]-ps[i]).y;});
            map<ll, int> sweep;
            for (int i : id) {
                for (auto it = sweep.lower_bound(-ps[i].y);
                     it != sweep.end();
                     sweep.erase(it++)) {
                    int j = it->second;
                    P d = ps[i] - ps[j];
                    if (d.y > d.x) break;
                    edges.push_back(Path(i, j, d.y + d.x));
                }
                sweep[-ps[i].y] = i;
            }
            for (P& p : ps) if (k & 1) p.x = -p.x; else
                swap(p.x, p.y);
        }
        return edges;
    }
};

```

3.9 Point in Poly

```

/*
 * Usage :
 * Returns true if p lies within the polygon. If strict is
 * true,
 * it returns false for points on the boundary.
 * The algorithm uses products in intermediate steps so watch
 * out for overflow.
 * Time complexity : O(n)
 */

template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
    int cnt = 0, n = sz(p);
    rep(i,0,n) {
        P q = p[(i + 1) % n];
        if (onSegment(p[i], q, a)) return !strict;
        //or: if (segDist(p[i], q, a) <= eps) return !strict;
        cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q) > 0;
    }
    return cnt;
}

```

3.10 Smallest Enclosing Circle

```

typedef Point<double> P;

struct Circle {
    P center;
    double radius;
};

```

```

Circle(P c, double r) : center(c), radius(r) {}
};

// Function to calculate the Euclidean distance between two
// points
double distance(const P& p1, const P& p2) {
    return hypot(p1.x - p2.x, p1.y - p2.y);
}

// Check if a point is inside a given circle
bool isPointInsideCircle(const P& p, const Circle& c) {
    return distance(p, c.center) <= c.radius;
}

// Create a circle from two points
Circle circleFromTwoPoints(const P& p1, const P& p2) {
    P center((p1.x + p2.x) / 2.0, (p1.y + p2.y) / 2.0);
    double radius = distance(p1, p2) / 2.0;
    return Circle(center, radius);
}

// Create a circle from three points using the circumcircle
// formula
Circle circleFromThreePoints(const P& p1, const P& p2, const
P& p3) {
    double ax = p1.x, ay = p1.y;
    double bx = p2.x, by = p2.y;
    double cx = p3.x, cy = p3.y;

    double d = 2 * (ax * (by - cy) + bx * (cy - ay) + cx * (ay
- by));
    if (d == 0) throw runtime_error("Collinear points");

    double ux = ((ax * ax + ay * ay) * (by - cy) + (bx * bx +
by * by) * (cy - ay) + (cx * cx + cy * cy) * (ay - by)) /
d;
    double uy = ((ax * ax + ay * ay) * (cx - bx) + (bx * bx +
by * by) * (ax - cx) + (cx * cx + cy * cy) * (bx - ax)) /
d;

    P center(ux, uy);
    double radius = distance(center, p1);
    return Circle(center, radius);
}

// Welzl's algorithm to find the minimum enclosing circle
Circle welzlAlgorithm(vector<P>& points, vector<P> boundary =
{}) {
    if (points.empty() || boundary.size() == 3) {
        if (boundary.empty()) return Circle(P(0, 0), 0);
        if (boundary.size() == 1) return Circle(boundary[0],
0);
        if (boundary.size() == 2) return
circleFromTwoPoints(boundary[0], boundary[1]);
        return circleFromThreePoints(boundary[0], boundary[1],
boundary[2]);
    }

    P p = points.back();
    points.pop_back();

    Circle d = welzlAlgorithm(points, boundary);

    if (isPointInsideCircle(p, d)) {
        points.push_back(p);
        return d;
    }

    boundary.push_back(p);
    Circle result = welzlAlgorithm(points, boundary);
    points.push_back(p);
    return result;
}

Circle findMinimumEnclosingCircle(vector<P>& points) {
    srand(time(0));
    random_shuffle(points.begin(), points.end());
    return welzlAlgorithm(points);
}

```

3.11 Minkowski Sum

```

/*
  Usage: minkowskiSum(a: convex polygon, b: convex polygon)
*/

template<class Point>
vector<Point> minkowskiSum(vector<Point> a, vector<Point> b) {
    rotate(begin(a), min_element(begin(a), end(a)), end(a));
    rotate(begin(b), min_element(begin(b), end(b)), end(b));
    int n = a.size(), m = b.size();
    vector<Point> h(n + m + 1); h[0] = a[0] + b[0];
    int t = 1;
    for (int i = 0, j = 0; i < n || j < m; ) {
        if (i == n) j++;
        else if (j == m) i++;
        else {
            Point pa = a[(i + 1) % n] - a[i], pb = b[(j + 1) % m] - b[j];
            auto cr = pa.cross(pb);
            if (cr >= 0) i++;
            if (cr <= 0) j++;
        }
        h[t++] = (a[i % n] + b[j % m]);
    }
    return {h.begin(), h.begin() + t - (t >= 2 && h[0] == h[t - 1])};
}

```

4 Graph

4.1 Block Cut Tree

```

/*
  Usage:
  - generate_block_cut_tree(): duh, also return number of vertices
  - cc[u]: vertices inside block u
  - sz[u]: number of vertices, -1 if is joint
*/

const int N = 1e5 + 69;
int n, m;
vector<int> graph[N], dfs_tree[N];

int num[N], low[N];
bool isJoint[N];
vector<int> st;
int dfs_cnt;
vector<vector<int>> bbc;

void tarjan(int u, int p){
    num[u] = low[u] = ++dfs_cnt;

    vector<int> child;
    for(int v: graph[u]) {
        if (v == p) {p = -1; continue;}
        if (num[v]){
            minimize(low[u], num[v]);
        }
        else{
            child.push_back(v);
            tarjan(v, u);
            minimize(low[u], low[v]);
            if (low[v] >= num[u]){
                isJoint[u] = true;
            }
        }
    }
    if (u == p) isJoint[u] = isJoint[u] && (child.size() >= 2);
    for(int v: child){
        if (isJoint[u] && low[v] >= num[u])
            dfs_tree[u].push_back(-v);
        else dfs_tree[u].push_back(v);
    }
}

void get_block(int u){
    st.push_back(u);
    for(int v: dfs_tree[u]){

```

```

        get_block(abs(v));
        if (v < 0){
            bbc.push_back({});
            while(bbc.back().empty() || bbc.back().back() != abs(v)){
                bbc.back().push_back(st.back());
                st.pop_back();
            }
            bbc.back().push_back(u);
        }
    }
}

vector<int> bc_tree[2*N];
int joint_cnt = 0;
int pos[2*N];
int sz[2 * N];
vector<int> cc[2 * N];

int generate_block_cut_tree(){
    dfs_cnt = 0;
    for(int i = 1; i <= n; ++i) if (num[i] == 0){
        tarjan(i, i);
        get_block(i);
        if (st.size()) {
            bbc.push_back(st);
            st.clear();
        }
    }

    int idx = 0;
    for(int i = 1; i <= n; ++i){
        if (!isJoint[i]) continue;
        joint_cnt++;
        pos[i] = ++idx;
        cc[idx].push_back(-i);
        sz[idx] = 1;
    }

    for(vector<int> i: bbc){
        ++idx;
        sz[idx] = i.size();
        cc[idx] = i;
        for(int j: i){
            if (isJoint[j]){
                int u = idx, v = pos[j];
                sz[v]--;
                bc_tree[u].push_back(v);
                bc_tree[v].push_back(u);
            }
        }
    }
    return idx;
}

```

4.2 Dominator Tree

```

struct DominatorTree {
    int n, id;
    vector<vector<int>> g, rg, bucket;
    vector<int> sdom, dom, vis, rev, pa, rt, mn, res;
    // dom[s] = s, dom[v] = -1 if s -> v not exists
    int query(int v, int x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], 1);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdom[mn[v]] > sdom[mn[rt[v]]])
            mn[v] = mn[rt[v]];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdom[id] = id, id++;
        for (int u : g[v]) {
            if (vis[u] == -1) dfs(u), pa[vis[u]] = vis[v];
            rg[vis[u]].pb(vis[v]);
        }
    }
    void build(int s) {
        dfs(s);
    }
}

```



```

    for (int i = id - 1; ~i; --i) {
        for (int u : rg[i])
            sdom[i] = min(sdom[i], sdom[query(u, 0)]);
        if (i) bucket[sdom[i]].pb(i);
        for (int u : bucket[i]) {
            int p = query(u, 0);
            dom[u] = sdom[p] == i ? i : p;
        }
        if (i) rt[i] = pa[i];
    }
    fill(all(res), -1);
    for (int i = 1; i < id; ++i) {
        if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
    }
    for (int i = 1; i < id; ++i)
        res[rev[i]] = rev[dom[i]];
    res[s] = s;
    for (int i = 0; i < n; ++i) dom[i] = res[i];
}
void add_edge(int u, int v) { g[u].pb(v); }
DominatorTree(int _n) : n(_n), id(0), g(n), rg(n),
    bucket(n), sdom(n), dom(n, -1), vis(n, -1),
    rev(n), pa(n), rt(n), mn(n), res(n) {
}
};

```

4.3 Eulerian Path

```

/*
Usage:
- dfs(u, p): dfs, what the fuck
- Return the order of edge you enter (in reverse)
*/

vector<int> ans;
void find_eulerian_path(int u) {
    while (graph[u].size()) {
        pair<int, int> v = graph[u].back();
        graph[u].pop_back();
        if (used[v.second]) continue;
        used[v.second] = true;
        find_eulerian_path(v.first);
        ans.push_back(v.first);
    }
}

```

4.4 Flow With Demand

```

/*
Usage:
- Dependency: Dinic template.
- FlowWithDemand(n, s, t): initialize a graph of size n, with
s and t being the source and sink nodes, respectively
- add_edge(u, v, d, w): add an edge from u to v, with
capacity w, and demand d.
- try_flow(capacity): verify whether there exists a flow,
such that the total residue is <= capacity.
*/

struct FlowWithDemand {
    int n, s, t, fake_s, fake_t;
    vector<long long> balance;
    Dinic dinic;

    FlowWithDemand(int n, int source, int sink)
        : n(n), s(source), t(sink), fake_s(n), fake_t(n + 1),
        balance(n, 0), dinic(n + 2, n, n + 1) {}

    void add_edge(int u, int v, long long demand, long long
    capacity) {
        balance[u] -= demand;
        balance[v] += demand;
        dinic.add_edge(u, v, capacity - demand);
    }

    bool try_flow(long long capacity) {
        vector<int> extras;
        for (int i = 0; i < n; ++i) {
            if (balance[i] > 0) {
                dinic.add_edge(fake_s, i, balance[i]);
                extras.push_back(fake_s);
            }
        }
    }
};

```

```

        extras.push_back(i);
    } else if (balance[i] < 0) {
        dinic.add_edge(i, fake_t, -balance[i]);
        extras.push_back(fake_t);
        extras.push_back(i);
    }
}
extras.push_back(s); extras.push_back(t);
dinic.add_edge(t, s, capacity);

bool valid = dinic.max_flow() >=
accumulate(balance.begin(), balance.end(), 0LL,
[] (long long sum, long long x) {
    return sum + max(0LL, x);
});

for (int i : extras) {
    dinic.adj[i].pop_back();
}

return valid;
}
};

```

4.5 General Max Matching

```

/*
Usage:
- Blossom(n): initialize a graph of size n
- add_edge(u, v): add an edge from u to v
- max_matching(): compute max matching
*/

struct Blossom {
    int N;
    vector<int> match, par, base, vis;
    vector<vector<int>>> adj;
    queue<int> Q;

    Blossom(int n) : N(n), match(n, -1), par(n, -1), base(n),
    vis(n, -1), adj(n) {
        iota(base.begin(), base.end(), 0);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    int find_lca(int u, int v) {
        vector<bool> visited(N);
        while (1) {
            u = base[u];
            visited[u] = true;
            if (match[u] == -1) break;
            u = par[match[u]];
        }
        while (1) {
            v = base[v];
            if (visited[v]) return v;
            v = par[match[v]];
        }
    }

    void blossom(int u, int v, int lca) {
        while (base[u] != lca) {
            par[u] = v;
            v = match[u];
            if (vis[v] == 1) Q.push(v), vis[v] = 0;
            base[u] = base[v] = lca;
            u = par[v];
        }
    }

    bool bfs(int root) {
        fill(vis.begin(), vis.end(), -1);
        Q = queue<int>();
        Q.push(root), vis[root] = 0;

        while (!Q.empty()) {
            int u = Q.front();

```

```

    Q.pop();
    for (int v : adj[u]) {
        if (vis[v] == -1) {
            vis[v] = 1, par[v] = u;
            if (match[v] == -1) {
                while (u != -1) {
                    int next_u = match[u];
                    match[u] = v, match[v] = u;
                    v = next_u, u = (v == -1) ? -1 :
                        par[v];
                }
                return true;
            }
            Q.push(match[v]), vis[match[v]] = 0;
        } else if (vis[v] == 0 && base[u] != base[v]) {
            int lca = find_lca(u, v);
            blossom(u, v, lca);
            blossom(v, u, lca);
        }
    }
}
return false;
}

int max_matching() {
    int res = 0;
    for (int u = 0; u < N; ++u)
        if (match[u] == -1 && bfs(u))
            ++res;
    return res;
}
};

```

4.6 Max Flow

```

/*
Usage:
- Dinic(n, s, t): initialize a graph of size n, with s and t
  being the source and sink nodes, respectively
- add_edge(u, v, w): add an edge from u to v, with capacity
  w.
- max_flow(): compute maximum flow
*/
struct Dinic {
    struct Edge { int to, rev; ll cap, flow; };
    int n, s, t;
    vector<vector<Edge>> adj;
    vector<int> level, ptr;

    Dinic(int n, int source, int sink) : n(n), s(source),
        t(sink), adj(n), level(n), ptr(n) {}

    void add_edge(int u, int v, ll cap) {
        adj[u].push_back({v, (int)adj[v].size(), cap, 0});
        adj[v].push_back({u, (int)adj[u].size() - 1, 0, 0});
    }

    bool bfs() {
        fill(level.begin(), level.end(), -1);
        level[s] = 0;
        queue<int> q({s});
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto &e : adj[u]) {
                if (e.cap - e.flow > 0 && level[e.to] == -1) {
                    level[e.to] = level[u] + 1;
                    q.push(e.to);
                }
            }
        }
        return level[t] != -1;
    }

    ll dfs(int u, ll flow) {
        if (u == t || flow == 0) return flow;
        for (; ptr[u] < (int)adj[u].size(); ++ptr[u]) {
            Edge &e = adj[u][ptr[u]];
            if (level[e.to] == level[u] + 1 && e.cap - e.flow
                > 0) {
                ll pushed = dfs(e.to, min(flow, e.cap -
                    e.flow));

```

```

                if (pushed > 0) {
                    e.flow += pushed;
                    adj[e.to][e.rev].flow -= pushed;
                    return pushed;
                }
            }
        }
        return 0;
    }

    void reset_flows() {
        for (auto &u_edges : adj) {
            for (auto &e : u_edges) {
                e.flow = 0; // Reset flow to 0
            }
        }
    }

    ll max_flow() {
        ll flow = 0;
        while (bfs()) {
            fill(ptr.begin(), ptr.end(), 0);
            while (ll pushed = dfs(s, LLONG_MAX)) {
                flow += pushed;
            }
            reset_flows();
        }
        return flow;
    }
};

```

4.7 Max Matching

```

/*
Usage:
- Matching(n, s, t): initialize a graph of size n
- add_edge(u, v): add an edge from u to v
- max_matching(): return maximum matching
*/
struct Matching {
    int n; vector<vector<int>> adj; vector<int> pairU, pairV,
        dist, color;
    Matching(int n) : n(n), adj(n + 1), pairU(n + 1), pairV(n
        + 1), dist(n + 1), color(n + 1, -1) {}

    void add_edge(int u, int v) { adj[u].push_back(v),
        adj[v].push_back(u); }

    bool bfs() {
        queue<int> q;
        for (int u = 1; u <= n; ++u)
            if (!pairU[u] && color[u] == 0) dist[u] = 0,
                q.push(u);
            else dist[u] = INT_MAX;
        dist[0] = INT_MAX;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int v : adj[u])
                if (dist[pairV[v]] == INT_MAX)
                    dist[pairV[v]] = dist[u] + 1,
                        q.push(pairV[v]);
        }
        return dist[0] != INT_MAX;
    }

    bool dfs(int u) {
        if (!u) return true;
        for (int v : adj[u])
            if (dist[pairV[v]] == dist[u] + 1 &&
                dfs(pairV[v]))
                return pairV[v] = u, pairU[u] = v, true;
            dist[u] = INT_MAX;
        return false;
    }

    vector<pair<int, int>> max_matching() {
        function<void(int)> color_graph = [&](int u) {
            for (int v : adj[u]) if (color[v] == -1) color[v]
                = 1 - color[u], color_graph(v);
        };
    }
};

```

```

    for (int u = 1; u <= n; ++u) if (color[u] == -1)
        color[u] = 0, color_graph(u);
    while (bfs()) for (int u = 1; u <= n; ++u) if
        (!pairU[u] && color[u] == 0) dfs(u);
    vector<pair<int, int>> res;
    for (int u = 1; u <= n; ++u) if (pairU[u])
        res.emplace_back(u, pairU[u]);
    return res;
}
};

```

4.8 Min Cost Flow

```

/*
Usage:
- MCMF(n, s, t): initialize a graph of size n, with s and t
  being the source and sink nodes, respectively
- add_edge(u, v, c, w): add an edge from u to v, with
  capacity c and cost w.
- minCostFlow(): compute {flow, min_cost}
*/

struct MCMF {
    struct Edge { int v, cap, cost, rev; };
    int N, s, t;
    vector<vector<Edge>> adj;
    vector<int> dist, parent, parentEdge, inQueue;

    MCMF(int n, int s, int t) : N(n), s(s), t(t), adj(n),
        dist(n), parent(n), parentEdge(n), inQueue(n) {}

    void addEdge(int u, int v, int cap, int cost) {
        adj[u].push_back({v, cap, cost, (int)adj[v].size()});
        adj[v].push_back({u, 0, -cost, (int)adj[u].size() - 1});
    }

    bool spfa(int s, int t) {
        fill(dist.begin(), dist.end(), INT_MAX);
        queue<int> q;
        dist[s] = 0; q.push(s); inQueue[s] = 1;
        while (!q.empty()) {
            int u = q.front(); q.pop(); inQueue[u] = 0;
            for (int i = 0; i < adj[u].size(); ++i) {
                Edge &e = adj[u][i];
                if (e.cap > 0 && dist[e.v] > dist[u] + e.cost) {
                    dist[e.v] = dist[u] + e.cost;
                    parent[e.v] = u; parentEdge[e.v] = i;
                    if (!inQueue[e.v]) q.push(e.v),
                        inQueue[e.v] = 1;
                }
            }
        }
        return dist[t] != INT_MAX;
    }

    pair<int, int> minCostFlow(int k) {
        int flow = 0, cost = 0;
        while (flow < k && spfa(s, t)) {
            int augFlow = k - flow;
            for (int u = t; u != s; u = parent[u])
                augFlow = min(augFlow,
                    adj[parent[u]][parentEdge[u]].cap);

            for (int u = t; u != s; u = parent[u]) {
                Edge &e = adj[parent[u]][parentEdge[u]];
                e.cap -= augFlow;
                adj[u][e.rev].cap += augFlow;
                cost += augFlow * e.cost;
            }

            flow += augFlow;
        }
        return flow == k ? make_pair(flow, cost) :
            make_pair(-1, -1);
    }
};

```

4.9 Tarjan

```
/*
```

Usage:

- dfs(u, p): dfs, what the fuck
- topo[u]: the strongly connected component that u belongs to, sorted topologically.

```
*/
```

```

int n, m;
vector<int> graph[N];
int num[N], low[N], topo[N];
deque<int> st;
int dfs_cnt = 0, topo_cnt = 0;

void dfs(int u, int p){
    num[u] = low[u] = ++dfs_cnt;
    st.push_back(u);
    int cnt = 0;
    for(int v: graph[u]){
        if (v == p && cnt == 0){cnt++; continue;}
        if (!num[v]){
            dfs(v, u);
            minimize(low[u], low[v]);
        }
        else minimize(low[u], num[v]);
    }

    if (num[u] == low[u]){
        topo_cnt++;
        while(st.back() != u){
            topo[st.back()] = topo_cnt;
            st.pop_back();
        }
        topo[u] = topo_cnt; st.pop_back();
    }
}

```

4.10 Two Sat

```

/*
Usage:
- TwoSAT(n): initialize a graph of size n
- addDisjunction(u, v): u and v = true
- solve(): check if it's correct
*/

struct TwoSAT {
    int n, time = 0, sccCount = 0;
    vector<vector<int>> adj;
    vector<int> low, ids, scc;
    vector<bool> onStack;
    vector<int> assignment, st;

    TwoSAT(int n) : n(n) {
        adj.resize(2 * n);
        low.assign(2 * n, -1);
        ids.assign(2 * n, -1);
        scc.assign(2 * n, -1);
        onStack.assign(2 * n, false);
        assignment.resize(n);
        st.reserve(2 * n);
    }

    inline void addDisjunction(int u, int v) {
        adj[u ^ 1].push_back(v);
        adj[v ^ 1].push_back(u);
    }

    void tarjanDFS(int v) {
        ids[v] = low[v] = time++;
        st.push_back(v);
        onStack[v] = true;

        for (int u : adj[v]) {
            if (ids[u] == -1)
                tarjanDFS(u);
            if (onStack[u])
                low[v] = min(low[v], low[u]);
        }

        if (ids[v] == low[v]) {
            while (true) {
                int u = st.back();

```

```

        st.pop_back();
        onStack[u] = false;
        scc[u] = sccCount;
        if (u == v) break;
    }
    ++sccCount;
}
}

bool solve() {
    for (int i = 0; i < 2 * n; ++i)
        if (ids[i] == -1)
            tarjanDFS(i);
    for (int i = 0; i < n; ++i) {
        if (scc[2 * i] == scc[2 * i + 1])
            return false;
        assignment[i] = (scc[2 * i] < scc[2 * i + 1]);
    }
    return true;
}
};

```

4.11 Flow Model

- Maximum/Minimum flow with lower bound / Circulation problem
 - Construct super source S and sink T .
 - For each edge (x, y, l, u) , connect $x \rightarrow y$ with capacity $u - l$.
 - For each vertex v , denote by $in(v)$ the difference between the sum of **incoming** lower bounds and the sum of **outgoing** lower bounds.
 - If $in(v) > 0$, connect $S \rightarrow v$ with capacity $in(v)$, otherwise, connect $v \rightarrow T$ with capacity $-in(v)$.
 - To maximize, connect $t \rightarrow s$ with capacity ∞ (skip this in circulation problem), and let f be the maximum flow from S to T . If $f \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, the maximum flow from s to t is the answer.
 - To minimize, let f be the maximum flow from S to T . Connect $t \rightarrow s$ with capacity ∞ and let the flow from S to T be f' . If $f + f' \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, f' is the answer.
 - The solution of each edge e is $l_e + f_e$, where f_e corresponds to the flow of edge e on the graph.
- Construct minimum vertex cover from maximum matching M on bipartite graph (X, Y)
 - Redirect every edge: $y \rightarrow x$ if $(x, y) \in M$, $x \rightarrow y$ otherwise.
 - DFS from unmatched vertices in X .
 - $x \in X$ is chosen iff x is unvisited.
 - $y \in Y$ is chosen iff y is visited.
- Minimum cost cyclic flow
 - Construct super source S and sink T
 - For each edge (x, y, c) , connect $x \rightarrow y$ with $(cost, cap) = (c, 1)$ if $c > 0$, otherwise connect $y \rightarrow x$ with $(cost, cap) = (-c, 1)$
 - For each edge with $c < 0$, sum these cost as K , then increase $d(y)$ by 1, decrease $d(x)$ by 1
 - For each vertex v with $d(v) > 0$, connect $S \rightarrow v$ with $(cost, cap) = (0, d(v))$
 - For each vertex v with $d(v) < 0$, connect $v \rightarrow T$ with $(cost, cap) = (0, -d(v))$
 - Flow from S to T , the answer is the cost of the flow $C + K$
- Maximum density induced subgraph
 - Binary search on answer, suppose we're checking answer T
 - Construct a max flow model, let K be the sum of all weights
 - Connect source $s \rightarrow v$, $v \in G$ with capacity K
 - For each edge (u, v, w) in G , connect $u \rightarrow v$ and $v \rightarrow u$ with capacity w
 - For $v \in G$, connect it with sink $v \rightarrow t$ with capacity $K + 2T - (\sum_{e \in E(v)} w(e)) - 2w(v)$

6. T is a valid answer if the maximum flow $f < K|V|$

- Minimum weight edge cover

- Change the weight of each edge to $\mu(u) + \mu(v) - w(u, v)$, where $\mu(v)$ is the cost of the cheapest edge incident to v .
- Let the maximum weight matching of the graph be x , the answer will be $\sum \mu(v) - x$.

5 String

5.1 Aho Corasick

```

/*
checked
*/

int nx[N][26];
int suf_link[N], par[N];
int leaf[N];
int num_node = 0;
int exlink[N];

int add(const string& s)
{
    int u = 0;
    for (const char& c : s)
    {
        if (nx[u][c - 'a'] == -1) nx[u][c - 'a'] = ++num_node;
        par[nx[u][c - 'a']] = u;
        u = nx[u][c - 'a'];
    }
    leaf[u] = 1;
    return u;
}

void build()
{
    queue<int> q;
    q.push(0);
    while (sz(q))
    {
        int u = q.front(); q.pop();
        int v = suf_link[u];
        if (leaf[v]) exlink[u] = v;
        else exlink[u] = exlink[v];
        for (int i = 0; i < 26; ++i)
        {
            int x = nx[u][i];
            if (nx[u][i] == -1)
                nx[u][i] = u == 0 ? 0 : nx[v][i];
            else
            {
                suf_link[x] = u == 0 ? 0 : nx[v][i];
                q.push(x);
            }
        }
    }
}
}

```

5.2 KMP

```

/*
KMP template
*/

struct KMP{
    struct Node{
        array<int, 2> nxt;
        int pi;
        Node(){nxt[0] = nxt[1] = 0; pi = -1;}
    };

    int n;
    vector<Node> a;

    KMP(string s){
        n = s.size();
        a.resize(n+1);
        s = "#" + s;
        for(int i = 0; i < n; ++i){

```

```

        if (i < n){
            a[i + 1].pi = a[i].nxt[s[i+1] - '0'];
            a[i].nxt[s[i+1] - '0'] = i + 1;
            a[i+1].nxt = a[a[i + 1].pi].nxt;
        }
    }
}

int count_matches(string s){
    if (s.size() < n) return 0;
    int x = 0;
    int ans = 0;
    for(char c: s){
        x = a[x].nxt[c - '0'];
        if (x == n) ans++;
    }
    return ans;
}
};

```

5.3 Palindrome Tree

```

/* THIS SHIT RUN IN O(N) TOO */

struct PalindromeTree{
    const static int K = 26;
    struct Node{
        long child[K], layer, pi;
        Node(long _layer = 0){
            memset(child, -1, sizeof child);
            layer = _layer;
            pi = -1;
        }
    };

    vector<Node> a;

    PalindromeTree(){
        a.push_back(Node(-1)); a.push_back(Node(0));
        a[1].pi = a[0].pi = 0;
    }

    void add_child(long &x, long layer){
        x = a.size();
        a.push_back(Node(layer));
    }

    void build(string s){
        long id = 0;
        s = "#" + s;
        for(long i = 1; i < s.size(); ++i){
            long digit = s[i] - '0';
            long j = id;
            while(j >= 0){
                if (s[i] == s[i - a[j].layer - 1]) break;
                j = a[j].pi;
            }

            if (a[j].child[digit] != -1) {
                id = a[j].child[digit];
                continue;
            }

            add_child(a[j].child[digit], a[j].layer + 2);
            long v = a[j].child[digit];

            if (a[v].layer == 1) j = 1;
            else if (a[v].layer == 2) j = a[0].child[digit];
            else{
                j = a[j].pi;
                while(j >= 0){
                    if (j == 0 || (s[i] == s[i - a[j].layer - 1])) break;
                    j = a[j].pi;
                }
                j = a[j].child[digit];
            }
            a[v].pi = j;
            id = v;
        }
        cout << a.size() - 2 << "\n";
    }
}

```

```

    }
};

```

5.4 Suffix Array

```

#define vi vector<int>
#define rep(i, l, r) for(int i = (l); i < (r); ++i)
struct SuffixArray {
    vi sa, lcp;
    SuffixArray(string s, int lim = 256) {
        // push 0 for 1-indexed and ("a" < "aa")
        // push 127 for 0-indexed and ("aa" < "a")
        s.push_back(0);
        int n = sz(s), k = 0, a, b;
        vi x(all(s)), y(n), ws(max(n, lim));
        sa = lcp = y, iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p)
        {
            p = j, iota(all(y), n - j);
            rep(i, 0, n) if (sa[i] >= j) y[p++] = sa[i] - j;
            fill(all(ws), 0);
            rep(i, 0, n) ws[x[i]]++;
            rep(i, 1, lim) ws[i] += ws[i - 1];
            for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
            swap(x, y), p = 1, x[sa[0]] = 0;
            rep(i, 1, n) a = sa[i - 1], b = sa[i], x[b] =
                (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
        }
        // lcp[i] = lcp(sa[i - 1], sa[i])
        for (int i = 0, j; i < n - 1; i++) {
            if (x[i] == 0) { k = 0; continue; }
            for (k && k--, j = sa[x[i] - 1]; s[i + k] == s[j + k]; k++);
            lcp[x[i]] = k;
        }
    }
};

```

5.5 Suffix Automaton

```

/* IT IS THE SHORTEST VER I CAN FIND */
// Time complexity : O(N)

struct SuffixAutomaton {
    vector<vector<int>> next;
    vector<int> len, link;
    int last, size;

    SuffixAutomaton(const string &s) {
        size = 1; last = 0;
        len.push_back(0); link.push_back(-1);
        next.push_back(vector<int>(26, -1));

        for (char c : s) {
            int cur = size++;
            len.push_back(len[last] + 1);
            next.push_back(vector<int>(26, -1));
            link.push_back(-1);

            int p = last;
            while (p != -1 && next[p][c - 'a'] == -1) {
                next[p][c - 'a'] = cur;
                p = link[p];
            }

            if (p == -1) link[cur] = 0;
            else {
                int q = next[p][c - 'a'];
                if (len[p] + 1 == len[q]) link[cur] = q;
                else {
                    int clone = size++;
                    len.push_back(len[p] + 1);
                    next.push_back(next[q]);
                    link.push_back(link[q]);
                    while (p != -1 && next[p][c - 'a'] == q) {
                        next[p][c - 'a'] = clone;
                        p = link[p];
                    }
                    link[q] = link[cur] = clone;
                }
            }
        }
    }
};

```

```

        last = cur;
    }
}
};

```

5.6 Z Function

```

/* YEP! IT IS JUST A Z FUNCTION */
// Time complexity : O(N)

vector<int> ZFunction(const string &S) {
    int n = S.length();
    vector<int> Z(n);
    for (int L = 0, R = 0, i = 1; i < n; ++i) {
        if (i <= R) Z[i] = min(R - i + 1, Z[i - L]);
        while (i + Z[i] < n && S[Z[i]] == S[i + Z[i]]) Z[i]++;
        if (i + Z[i] - 1 > R) L = i, R = i + Z[i] - 1;
    }
    Z[0] = n;
    return Z;
}

```

6 Tree

6.1 Tree Line

```

/*
Usage: that's the dsu, push the edges in, and get the
chains. Now you have the treeline
*/

```

```

struct DSU{
    int n;
    vector<int> parent, sz;
    vector<vector<pair<int, int>>> chain;

    DSU(int _n){
        n = _n;
        parent.resize(n+1); sz.resize(n+1, 1);
        chain.resize(n+1);
        for(int i = 0; i<=n; ++i) {
            parent[i] = i;
            chain[i] = {{i, 0}};
        }

        int find_set(int u){return (u == parent[u]) ? u :
        (parent[u] = find_set(parent[u]));}
        bool same_set(int u, int v){return find_set(u) ==
        find_set(v);}

        bool join_set(int u, int v, int w){
            u = find_set(u), v = find_set(v);
            if (u != v){
                if (sz[u] < sz[v]) swap(u, v);
                parent[v] = u;
                sz[u] += sz[v];
                chain[v][0].second = w;
                for(pair<int, int> i: chain[v])
                    chain[u].push_back(i);
                return true;
            }
            return false;
        }

        int get_size(int u){return sz[find_set(u)];}
};

```

7 Arithmetic

7.1 Big Num Long

```

/* BIGNUM AGAIN!! BUT TESTED BY I_LOVE_HOANG_YEN */
// BigInt {
const int BASE_DIGITS = 9;
const int BASE = 1000000000;

struct BigInt {

```

```

    int sign;
    vector<int> a;

    // ----- Constructors -----
    // Default constructor.
    BigInt() : sign(1) {}

    // Constructor from long long.
    BigInt(long long v) {
        *this = v;
    }

    BigInt& operator = (long long v) {
        sign = 1;
        if (v < 0) {
            sign = -1;
            v = -v;
        }
        a.clear();
        for (; v > 0; v = v / BASE)
            a.push_back(v % BASE);
        return *this;
    }

    // Initialize from string.
    BigInt(const string& s) {
        read(s);
    }

    // ----- Input / Output -----
    void read(const string& s) {
        sign = 1;
        a.clear();
        int pos = 0;
        while (pos < (int) s.size() && (s[pos] == '-' ||
        s[pos] == '+')) {
            if (s[pos] == '-')
                sign = -sign;
            ++pos;
        }
        for (int i = s.size() - 1; i >= pos; i -= BASE_DIGITS) {
            int x = 0;
            for (int j = max(pos, i - BASE_DIGITS + 1); j <=
            i; j++)
                x = x * 10 + s[j] - '0';
            a.push_back(x);
        }
        trim();
    }

    friend istream& operator>>(istream &stream, BigInt &v) {
        string s;
        stream >> s;
        v.read(s);
        return stream;
    }

    friend ostream& operator<<(ostream &stream, const BigInt
    &v) {
        if (v.sign == -1 && !v.isZero())
            stream << '-';
        stream << (v.a.empty() ? 0 : v.a.back());
        for (int i = (int) v.a.size() - 2; i >= 0; --i)
            stream << setw(BASE_DIGITS) << setfill('0') <<
            v.a[i];
        return stream;
    }

    // ----- Comparison -----
    bool operator<(const BigInt &v) const {
        if (sign != v.sign)
            return sign < v.sign;
        if (a.size() != v.a.size())
            return a.size() * sign < v.a.size() * v.sign;
        for (int i = ((int) a.size()) - 1; i >= 0; i--)
            if (a[i] != v.a[i])
                return a[i] * sign < v.a[i] * v.sign;
        return false;
    }

    bool operator>(const BigInt &v) const {

```



```

        return v < *this;
    }
    bool operator<=(const BigInt &v) const {
        return !(v < *this);
    }
    bool operator>=(const BigInt &v) const {
        return !(*this < v);
    }
    bool operator==(const BigInt &v) const {
        return !(*this < v) && !(v < *this);
    }
    bool operator!=(const BigInt &v) const {
        return *this < v || v < *this;
    }

    // Returns:
    // 0 if |x| == |y|
    // -1 if |x| < |y|
    // 1 if |x| > |y|
    friend int __compare_abs(const BigInt& x, const BigInt& y)
    {
        if (x.a.size() != y.a.size()) {
            return x.a.size() < y.a.size() ? -1 : 1;
        }

        for (int i = ((int) x.a.size()) - 1; i >= 0; --i) {
            if (x.a[i] != y.a[i]) {
                return x.a[i] < y.a[i] ? -1 : 1;
            }
        }
        return 0;
    }

    // ----- Unary operator - and operators +-
    -----
    BigInt operator-() const {
        BigInt res = *this;
        if (isZero()) return res;

        res.sign = -sign;
        return res;
    }

    // Note: sign ignored.
    void __internal_add(const BigInt& v) {
        if (a.size() < v.a.size()) {
            a.resize(v.a.size(), 0);
        }
        for (int i = 0, carry = 0; i < (int) max(a.size(),
            v.a.size()) || carry; ++i) {
            if (i == (int) a.size()) a.push_back(0);

            a[i] += carry + (i < (int) v.a.size() ? v.a[i] :
                0);
            carry = a[i] >= BASE;
            if (carry) a[i] -= BASE;
        }
    }

    // Note: sign ignored.
    void __internal_sub(const BigInt& v) {
        for (int i = 0, carry = 0; i < (int) v.a.size() ||
            carry; ++i) {
            a[i] -= carry + (i < (int) v.a.size() ? v.a[i] :
                0);
            carry = a[i] < 0;
            if (carry) a[i] += BASE;
        }
        this->trim();
    }

    BigInt operator += (const BigInt& v) {
        if (sign == v.sign) {
            __internal_add(v);
        } else {
            if (__compare_abs(*this, v) >= 0) {
                __internal_sub(v);
            } else {
                BigInt vv = v;
                swap(*this, vv);
                __internal_sub(vv);
            }
        }
    }

```

```

    }
    return *this;
}

BigInt operator -= (const BigInt& v) {
    if (sign == v.sign) {
        if (__compare_abs(*this, v) >= 0) {
            __internal_sub(v);
        } else {
            BigInt vv = v;
            swap(*this, vv);
            __internal_sub(vv);
            this->sign = -this->sign;
        }
    } else {
        __internal_add(v);
    }
    return *this;
}

template< typename L, typename R >
typename std::enable_if<
    std::is_convertible<L, BigInt>::value &&
    std::is_convertible<R, BigInt>::value &&
    std::is_lvalue_reference<R&&>::value,
    BigInt>::type friend operator + (L&& l, R&& r) {
    BigInt result(std::forward<L>(l));
    result += r;
    return result;
}

template< typename L, typename R >
typename std::enable_if<
    std::is_convertible<L, BigInt>::value &&
    std::is_convertible<R, BigInt>::value &&
    std::is_rvalue_reference<R&&>::value,
    BigInt>::type friend operator + (L&& l, R&& r) {
    BigInt result(std::move(r));
    result += l;
    return result;
}

template< typename L, typename R >
typename std::enable_if<
    std::is_convertible<L, BigInt>::value &&
    std::is_convertible<R, BigInt>::value &&
    std::is_lvalue_reference<R&&>::value,
    BigInt>::type friend operator - (L&& l, R&& r) {
    BigInt result(std::forward<L>(l));
    result -= r;
    return result;
}

// ----- Operators * / %
-----
friend pair<BigInt, BigInt> divmod(const BigInt& a1, const
    BigInt& b1) {
    assert(b1 > 0); // divmod not well-defined for b < 0.

    long long norm = BASE / (b1.a.back() + 1);
    BigInt a = a1.abs() * norm;
    BigInt b = b1.abs() * norm;
    BigInt q = 0, r = 0;
    q.a.resize(a.a.size());

    for (int i = a.a.size() - 1; i >= 0; i--) {
        r *= BASE;
        r += a.a[i];
        long long s1 = r.a.size() <= b.a.size() ? 0 :
            r.a[b.a.size()];
        long long s2 = r.a.size() <= b.a.size() - 1 ? 0 :
            r.a[b.a.size() - 1];
        long long d = ((long long) BASE * s1 + s2) /
            b.a.back();
        r -= b * d;
        while (r < 0) {
            r += b, --d;
        }
        q.a[i] = d;
    }

    q.sign = a1.sign * b1.sign;
    r.sign = a1.sign;
}

```

```

    q.trim();
    r.trim();
    auto res = make_pair(q, r / norm);
    if (res.second < 0) res.second += b1;
    return res;
}

BigInt operator/(const BigInt &v) const {
    if (v < 0) return divmod(*this, -v).first;
    return divmod(*this, v).first;
}

BigInt operator%(const BigInt &v) const {
    return divmod(*this, v).second;
}

void operator/=(int v) {
    assert(v > 0); // operator / not well-defined for v
    <= 0.
    if (llabs(v) >= BASE) {
        *this /= BigInt(v);
        return;
    }
    if (v < 0)
        sign = -sign, v = -v;
    for (int i = (int) a.size() - 1, rem = 0; i >= 0; --i) {
        long long cur = a[i] + rem * (long long) BASE;
        a[i] = (int) (cur / v);
        rem = (int) (cur % v);
    }
    trim();
}

BigInt operator/(int v) const {
    assert(v > 0); // operator / not well-defined for v
    <= 0.

    if (llabs(v) >= BASE) {
        return *this / BigInt(v);
    }
    BigInt res = *this;
    res /= v;
    return res;
}

void operator/=(const BigInt &v) {
    *this = *this / v;
}

long long operator%(long long v) const {
    assert(v > 0); // operator / not well-defined for v
    <= 0.
    assert(v < BASE);
    int m = 0;
    for (int i = a.size() - 1; i >= 0; --i)
        m = (a[i] + m * (long long) BASE) % v;
    return m * sign;
}

void operator*=(int v) {
    if (llabs(v) >= BASE) {
        *this *= BigInt(v);
        return;
    }
    if (v < 0)
        sign = -sign, v = -v;
    for (int i = 0, carry = 0; i < (int) a.size() ||
        carry; ++i) {
        if (i == (int) a.size())
            a.push_back(0);
        long long cur = a[i] * (long long) v + carry;
        carry = (int) (cur / BASE);
        a[i] = (int) (cur % BASE);
    }
    trim();
}

BigInt operator*(int v) const {
    if (llabs(v) >= BASE) {
        return *this * BigInt(v);
    }
    BigInt res = *this;
    res *= v;

```

```

    return res;
}

// Convert BASE 10^old --> 10^new.
static vector<int> convert_base(const vector<int> &a, int
old_digits, int new_digits) {
    vector<long long> p(max(old_digits, new_digits) + 1);
    p[0] = 1;
    for (int i = 1; i < (int) p.size(); ++i)
        p[i] = p[i - 1] * 10;
    vector<int> res;
    long long cur = 0;
    int cur_digits = 0;
    for (int i = 0; i < (int) a.size(); ++i) {
        cur += a[i] * p[cur_digits];
        cur_digits += old_digits;
        while (cur_digits >= new_digits) {
            res.push_back((long long) (cur %
            p[new_digits]));
            cur /= p[new_digits];
            cur_digits -= new_digits;
        }
    }
    res.push_back((int) cur);
    while (!res.empty() && !res.back())
        res.pop_back();
    return res;
}

void fft(vector<complex<double>> &x, bool invert) const {
    int n = (int) x.size();

    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n >> 1;
        for (; j >= bit; bit >>= 1)
            j -= bit;
        j += bit;
        if (i < j)
            swap(x[i], x[j]);
    }

    for (int len = 2; len <= n; len <= 1) {
        double ang = 2 * 3.14159265358979323846 / len *
        (invert ? -1 : 1);
        complex<double> wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            complex<double> w(1);
            for (int j = 0; j < len / 2; ++j) {
                complex<double> u = x[i + j];
                complex<double> v = x[i + j + len / 2] *
                w;
                x[i + j] = u + v;
                x[i + j + len / 2] = u - v;
                w *= wlen;
            }
        }
    }
    if (invert)
        for (int i = 0; i < n; ++i)
            x[i] /= n;
}

void multiply_fft(const vector<int> &x, const vector<int>
&y, vector<int> &res) const {
    vector<complex<double>> fa(x.begin(), x.end());
    vector<complex<double>> fb(y.begin(), y.end());
    int n = 1;
    while (n < (int) max(x.size(), y.size()))
        n <= 1;
    n <= 1;
    fa.resize(n);
    fb.resize(n);

    fft(fa, false);
    fft(fb, false);
    for (int i = 0; i < n; ++i)
        fa[i] *= fb[i];
    fft(fa, true);

    res.resize(n);
    long long carry = 0;
    for (int i = 0; i < n; ++i) {

```

```

        long long t = (long long) (fa[i].real() + 0.5) +
        carry;
        carry = t / 1000;
        res[i] = t % 1000;
    }
}

BigInt mul_simple(const BigInt &v) const {
    BigInt res;
    res.sign = sign * v.sign;
    res.a.resize(a.size() + v.a.size());
    for (int i = 0; i < (int) a.size(); ++i)
        if (a[i])
            for (int j = 0, carry = 0; j < (int)
                v.a.size() || carry; ++j) {
                long long cur = res.a[i + j] + (long long)
                a[i] * (j < (int) v.a.size() ? v.a[j] : 0)
                + carry;
                carry = (int) (cur / BASE);
                res.a[i + j] = (int) (cur % BASE);
            }
    res.trim();
    return res;
}

typedef vector<long long> vll;

static vll karatsubaMultiply(const vll &a, const vll &b) {
    int n = a.size();
    vll res(n + n);
    if (n <= 32) {
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++)
                res[i + j] += a[i] * b[j];
        return res;
    }

    int k = n >> 1;
    vll a1(a.begin(), a.begin() + k);
    vll a2(a.begin() + k, a.end());
    vll b1(b.begin(), b.begin() + k);
    vll b2(b.begin() + k, b.end());

    vll a1b1 = karatsubaMultiply(a1, b1);
    vll a2b2 = karatsubaMultiply(a2, b2);

    for (int i = 0; i < k; i++)
        a2[i] += a1[i];
    for (int i = 0; i < k; i++)
        b2[i] += b1[i];

    vll r = karatsubaMultiply(a2, b2);
    for (int i = 0; i < (int) a1b1.size(); i++)
        r[i] -= a1b1[i];
    for (int i = 0; i < (int) a2b2.size(); i++)
        r[i] -= a2b2[i];

    for (int i = 0; i < (int) r.size(); i++)
        res[i + k] += r[i];
    for (int i = 0; i < (int) a1b1.size(); i++)
        res[i] += a1b1[i];
    for (int i = 0; i < (int) a2b2.size(); i++)
        res[i + n] += a2b2[i];
    return res;
}

BigInt mul_karatsuba(const BigInt &v) const {
    vector<int> x6 = convert_base(this->a, BASE_DIGITS,
    6);
    vector<int> y6 = convert_base(v.a, BASE_DIGITS, 6);
    vll x(x6.begin(), x6.end());
    vll y(y6.begin(), y6.end());
    while (x.size() < y.size())
        x.push_back(0);
    while (y.size() < x.size())
        y.push_back(0);
    while (x.size() & (x.size() - 1))
        x.push_back(0), y.push_back(0);
    vll c = karatsubaMultiply(x, y);
    BigInt res;
    res.sign = sign * v.sign;
    long long carry = 0;

    for (int i = 0; i < (int) c.size(); i++) {
        long long cur = c[i] + carry;
        res.a.push_back((int) (cur % 1000000));
        carry = cur / 1000000;
    }
    res.a = convert_base(res.a, 6, BASE_DIGITS);
    res.trim();
    return res;
}

void operator*=(const BigInt &v) {
    *this = *this * v;
}

BigInt operator*(const BigInt &v) const {
    if (a.size() * v.a.size() <= 1000111) return
    mul_simple(v);
    if (a.size() > 500111 || v.a.size() > 500111) return
    mul_fft(v);
    return mul_karatsuba(v);
}

BigInt mul_fft(const BigInt& v) const {
    BigInt res;
    res.sign = sign * v.sign;
    multiply_fft(convert_base(a, BASE_DIGITS, 3),
    convert_base(v.a, BASE_DIGITS, 3), res.a);
    res.a = convert_base(res.a, 3, BASE_DIGITS);
    res.trim();
    return res;
}

// ----- Misc -----
BigInt abs() const {
    BigInt res = *this;
    res.sign *= res.sign;
    return res;
}

void trim() {
    while (!a.empty() && !a.back())
        a.pop_back();
    if (a.empty())
        sign = 1;
}

bool isZero() const {
    return a.empty() || (a.size() == 1 && !a[0]);
}

friend BigInt gcd(const BigInt &x, const BigInt &y) {
    return y.isZero() ? x : gcd(y, x % y);
}

friend BigInt lcm(const BigInt &x, const BigInt &y) {
    return x / gcd(x, y) * y;
}

friend BigInt sqrt(const BigInt &a1) {
    BigInt a = a1;
    while (a.a.empty() || a.a.size() % 2 == 1)
        a.a.push_back(0);

    int n = a.a.size();

    int firstDigit = (int) sqrt((double) a.a[n - 1] * BASE
    + a.a[n - 2]);
    int norm = BASE / (firstDigit + 1);
    a *= norm;
    a *= norm;
    while (a.a.empty() || a.a.size() % 2 == 1)
        a.a.push_back(0);

    BigInt r = (long long) a.a[n - 1] * BASE + a.a[n - 2];
    firstDigit = (int) sqrt((double) a.a[n - 1] * BASE +
    a.a[n - 2]);
    int q = firstDigit;
    BigInt res;

    for (int j = n / 2 - 1; j >= 0; j--) {
        for (; --q) {
            BigInt r1 = (r - (res * 2 * BigInt(BASE) + q)
            * q) * BigInt(BASE) * BigInt(BASE) + (j > 0 ?
            (long long) a.a[2 * j - 1] * BASE + a.a[2 * j
            - 2] : 0);

```

```

        if (r1 >= 0) {
            r = r1;
            break;
        }
    }
    res *= BASE;
    res += q;

    if (j > 0) {
        int d1 = res.a.size() + 2 < r.a.size() ?
            r.a[res.a.size() + 2] : 0;
        int d2 = res.a.size() + 1 < r.a.size() ?
            r.a[res.a.size() + 1] : 0;
        int d3 = res.a.size() < r.a.size() ?
            r.a[res.a.size()] : 0;
        q = ((long long) d1 * BASE * BASE + (long
            long) d2 * BASE + d3) / (firstDigit * 2);
    }

    res.trim();
    return res / norm;
}
};
// }}}

```

7.2 Chinese Remainder Theorem

```

tuple<int64_t, int64_t, int64_t> extended_gcd(int64_t a,
int64_t b) {
    if (b == 0) {
        return {a, 1, 0};
    }
    int64_t gcd, x1, y1;
    tie(gcd, x1, y1) = extended_gcd(b, a % b);
    return {gcd, y1, x1 - (a / b) * y1};
}

pair<int64_t, int64_t> chinese_remainder_theorem(int64_t a1,
int64_t m1, int64_t a2, int64_t m2) {
    int64_t gcd, x, y;
    std::tie(gcd, x, y) = extended_gcd(m1, m2);

    if ((a2 - a1) % gcd != 0) {
        return {0, -1};
    }

    int64_t lcm = (m1 / gcd) * m2;
    int64_t combined_solution = a1 + ((a2 - a1) / gcd * x %
        (m2 / gcd)) * m1;
    combined_solution = (combined_solution % lcm + lcm) % lcm;

    return {combined_solution, lcm};
}

```

7.3 Diophantine

```

int gcd(int a, int b, int& x, int& y) {
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    int x1, y1;
    int d = gcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

bool find_any_solution(int a, int b, int c, int &x0, int &y0,
int &g) {
    g = gcd(abs(a), abs(b), x0, y0);
    if (c % g) {
        return false;
    }

    x0 *= c / g;
    y0 *= c / g;
    if (a < 0) x0 = -x0;
    if (b < 0) y0 = -y0;
    return true;
}

```

```

}

void shift_solution(int &x, int &y, int a, int b, int cnt) {
    x += cnt * b;
    y -= cnt * a;
}

int find_all_solutions(int a, int b, int c, int minx, int
maxx, int miny, int maxy) {
    int x, y, g;
    if (!find_any_solution(a, b, c, x, y, g))
        return 0;
    a /= g;
    b /= g;

    int sign_a = a > 0 ? +1 : -1;
    int sign_b = b > 0 ? +1 : -1;

    shift_solution(x, y, a, b, (minx - x) / b);
    if (x < minx)
        shift_solution(x, y, a, b, sign_b);
    if (x > maxx)
        return 0;
    int lx1 = x;

    shift_solution(x, y, a, b, (maxx - x) / b);
    if (x > maxx)
        shift_solution(x, y, a, b, -sign_b);
    int rx1 = x;

    shift_solution(x, y, a, b, -(miny - y) / a);
    if (y < miny)
        shift_solution(x, y, a, b, -sign_a);
    if (y > maxy)
        return 0;
    int lx2 = x;

    shift_solution(x, y, a, b, -(maxy - y) / a);
    if (y > maxy)
        shift_solution(x, y, a, b, sign_a);
    int rx2 = x;

    if (lx2 > rx2)
        swap(lx2, rx2);
    int lx = max(lx1, lx2);
    int rx = min(rx1, rx2);

    if (lx > rx)
        return 0;
    return (rx - lx) / abs(b) + 1;
}

```

7.4 Dirichlet Convolution

```

struct Dirichlet
{
    typedef ll(*func)(ll);
    ll n, lim, inv;
    func p_f, p_g, p_fg;
    unordered_map<ll, ll> mem;
    Dirichlet() {}
    Dirichlet(func x, func y, func z) : p_f(x), p_g(y), p_fg(z)
    {}
    ll calc(ll x)
    {
        if (x <= lim) return p_f(x);
        auto d = mem.find(x);
        if (d != mem.end()) return d->second;
        ll ans = 0;
        for (ll i = 2, la; i <= x; i = la + 1)
        {
            la = x / (x / i);
            ans = ans + (p_g(la) - p_g(i - 1) + MOD) * calc(x
                / i) % MOD;
            if (ans >= MOD) ans -= MOD;
        }
        ans %= MOD;
        ans = p_fg(x) - ans;
        if (ans < 0) ans += MOD;
        ans = 1ll * ans * inv % MOD;
    }
}

```

```

        return mem[x] = ans;
    }
    ll solve(ll n_, ll lim_)
    {
        n = n_, lim = lim_;
        inv = bin_pow(p_g(1), MOD - 2);
        return calc(n);
    }
};

```

7.5 Extended GCD

```

int extendedGCD(int a, int b, int &x, int &y){
    x = 1, y = 0;
    int _x = 0, _y = 1;
    while(b > 0){
        int q = a / b;
        tie(x, _x) = make_tuple(_x, x - q * _x);
        tie(y, _y) = make_tuple(_y, y - q * _y);
        tie(a, b) = make_tuple(b, a - q * b);
    }
    return a;
}

```

7.6 Gaussian Elimination (Float)

```

/* GAUSS ELIMINATION YAHOOO! */

void gauss(vector<vector<double>>& a, vector<double>& b) {
    int n = a.size();
    for (int i = 0; i < n; ++i) {
        for (int j = i + 1; j < n; ++j) {
            if (a[j][i]) {
                double ratio = a[j][i] / a[i][i];
                for (int k = i; k < n; ++k) a[j][k] -= ratio * a[i][k];
                b[j] -= ratio * b[i];
            }
        }
    }
    for (int i = n - 1; i >= 0; --i) {
        for (int j = i + 1; j < n; ++j) b[i] -= a[i][j] * b[j];
        b[i] /= a[i][i];
    }
}

```

7.7 Gaussian Elimination (Integer)

```

const int MOD = 1e9 + 9;

ll fast_pow(ll a, ll n){
    ll ans = 1;
    while(n > 0){
        if (n & 1) ans = ans * a % MOD;
        a = a * a % MOD;
        n /= 2;
    }
    return ans;
}

ll inverse(ll a){return fast_pow(a, MOD - 2);}

ll normie(ll x){
    x %= MOD;
    if (x < 0) x += MOD;
    return x;
}

vector<ll> solve(vector<vector<ll>> sigma){
    int n = sigma.size(), m = sigma[0].size() - 1;
    vector<bool> banned(n);
    for(int j = 0; j < m; ++j){
        int idx = -1;
        for(int i = 0; i < n; ++i) if (!banned[i] && sigma[i][j]) {
            idx = i;
            ll head = sigma[i][j];
            ll inv = inverse(head);

```

```

            for(int k = 0; k <= m; ++k) sigma[i][k] =
                (sigma[i][k] * inv) % MOD;
        }
        if (idx == -1) continue;
        banned[idx] = true;
        for(int i = 0; i < n; ++i) if (sigma[i][j] && i != idx){
            for(int k = 0; k <= m; ++k) sigma[i][k] =
                normie(sigma[i][k] - sigma[idx][k]);
        }
    }

    vector<int> deg(n);
    for(int i = 0; i < n; ++i){
        deg[i] = m;
        for(int j = 0; j < m; ++j) if (sigma[i][j]){
            deg[i] = j;
            break;
        }
    }
    for(int i = 0; i < n; ++i){
        if (deg[i] == m) continue;
        while(true){
            int v = m;
            for(int j = deg[i] + 1; j < m; ++j) if
                (sigma[i][j]){
                    v = j;
                    break;
                }
            if (v == m) break;
            for(int x = 0; x < n; ++x) if (deg[x] == v){
                ll shit = inverse(sigma[x][v]) * sigma[i][v] %
                    MOD;
                for(int j = 0; j <= m; ++j) {
                    sigma[i][j] = normie(sigma[i][j] - shit *
                        sigma[x][j]);
                }
                break;
            }
        }
    }
    vector<ll> ans(m);
    for(int i = 0; i < n; ++i) {
        ans[deg[i]] = sigma[i].back();
    }
    return ans;
}

```

7.8 Lagrange Interpol

```

typedef long long ll;

// Function to compute (base^exp) % mod
ll mod_exp(ll base, ll exp, ll mod) {
    ll res = 1;
    base %= mod;
    while (exp > 0) {
        if (exp & 1) res = res * base % mod;
        base = base * base % mod;
        exp >>= 1;
    }
    return res;
}

// Function to compute modular inverse using Fermat's Little Theorem
ll mod_inv(ll a, ll mod) {
    return mod_exp(a, mod - 2, mod);
}

// Function to add two polynomials
std::vector<ll> add_polynomials(const std::vector<ll>& a,
    const std::vector<ll>& b) {
    std::vector<ll> result(std::max(a.size(), b.size()));
    for (size_t i = 0; i < result.size(); ++i) {
        if (i < a.size()) result[i] = (result[i] + a[i]) %
            MOD;
        if (i < b.size()) result[i] = (result[i] + b[i]) %
            MOD;
    }
    return result;
}

```

```
// Function to multiply a polynomial by a monomial (x - c)
std::vector<ll> multiply_polynomial(const std::vector<ll>&
poly, ll c) {
    std::vector<ll> result(poly.size() + 1);
    for (size_t i = 0; i < poly.size(); ++i) {
        result[i + 1] = poly[i];
        result[i] = (result[i] - c * poly[i] % MOD + MOD) %
MOD;
    }
    return result;
}

// Function to multiply a polynomial by a constant
std::vector<ll> multiply_by_constant(const std::vector<ll>&
poly, ll constant) {
    std::vector<ll> result(poly.size());
    for (size_t i = 0; i < poly.size(); ++i) {
        result[i] = poly[i] * constant % MOD;
    }
    return result;
}

// Function for Lagrange interpolation to find the polynomial
std::vector<ll> lagrange_polynomial(const std::vector<ll>&
x_vals, const std::vector<ll>& y_vals) {
    int n = x_vals.size();
    std::vector<ll> poly(n, 0);

    for (int i = 0; i < n; ++i) {
        ll denom = 1;
        std::vector<ll> term = {y_vals[i]};
        for (int j = 0; j < n; ++j) {
            if (i != j) {
                denom = denom * (x_vals[i] - x_vals[j] + MOD)
% MOD;
                term = multiply_polynomial(term, x_vals[j]);
            }
        }
        ll inv_denom = mod_inv(denom, MOD);
        term = multiply_by_constant(term, inv_denom);
        poly = add_polynomials(poly, term);
    }

    return poly;
}
```

7.9 Miller Rabin

```
namespace MillerRabin{
    using u64 = uint64_t;
    using u128 = __uint128_t;

    u64 binpower(u64 base, u64 e, u64 mod) {
        u64 result = 1;
        base %= mod;
        while (e) {
            if (e & 1)
                result = (u128)result * base % mod;
            base = (u128)base * base % mod;
            e >>= 1;
        }
        return result;
    }

    bool check_composite(u64 n, u64 a, u64 d, int s) {
        u64 x = binpower(a, d, n);
        if (x == 1 || x == n - 1)
            return false;
        for (int r = 1; r < s; r++) {
            x = (u128)x * x % n;
            if (x == n - 1)
                return false;
        }
        return true;
    };

    bool MillerRabin(u64 n) { // returns true if n is prime,
        else returns false.
        if (n < 2)
            return false;

```

```
int r = 0;
u64 d = n - 1;
while ((d & 1) == 0) {
    d >>= 1;
    r++;
}

for (int a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
37}) {
    if (n == a)
        return true;
    if (check_composite(n, a, d, r))
        return false;
}
return true;
}
```

7.10 Pollard Rho

```
// checked
ll mul(ll x, ll y, ll p) { return (x * y - (ll)((long double)x
/ p * y) * p + p) % p; } // __int128
vector<ll> chk = { 2, 325, 9375, 28178, 450775, 9780504,
1795265022 };
ll Pow(ll a, ll b, ll n) {
    ll res = 1;
    for (; b; b >>= 1, a = mul(a, a, n))
        if (b & 1) res = mul(res, a, n);
    return res;
}

bool check(ll a, ll d, int s, ll n) {
    a = Pow(a, d, n);
    if (a <= 1) return 1;
    for (int i = 0; i < s; ++i, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}

bool IsPrime(ll n) {
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d >>= 1, s++;
    for (ll i : chk) if (!check(i, d, s, n)) return 0;
    return 1;
} // 5761f3

const vector<ll> small = { 2, 3, 5, 7, 11, 13, 17, 19 };
ll FindFactor(ll n) {
    if (IsPrime(n)) return 1;
    for (ll p : small) if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {return (mul(a, a, n) + t) % n; };
    for (int l = 2; ; l <= 1) {
        x = y;
        int m = min(l, 32);
        for (int i = 0; i < l; i += m) {
            d = 1;
            for (int j = 0; j < m; ++j) {
                y = f(y), d = mul(d, abs(x - y), n);
            }
            ll g = __gcd(d, n);
            if (g == n) {
                l = 1, y = 2, t++;
                break;
            }
            if (g != 1) return g;
        }
    }
}

map<ll, int> res;
void PollardRho(ll n) {
    if (n == 1) return;
    if (IsPrime(n)) return res[n]++, void(0);
    ll d = FindFactor(n);
    PollardRho(n / d), PollardRho(d);
} // 57e9e3
```

7.11 Pythagorean Triples

```
subsubsection* { Pythagorean Triples }
```


The Pythagorean triples are uniquely generated by

```
[
a = k * (m^2 - n^2), quad b = k * (2mn), quad c
= k * (m^2 + n^2),
]
with m > n > 0, k > 0, m perp n, and either m or
n even.
```

7.12 Big Num Short

```
/*----- Big num template start
-----*/

string add(string a, string b){
    if (a.size() > b.size()) b = string(a.size() - b.size(),
    '0') + b;
    if (a.size() < b.size()) a = string(b.size() - a.size(),
    '0') + a;
    long carry = 0;
    for(long i = a.size() - 1; i>=0; --i){
        a[i] += b[i] - '0' + carry;
        if (a[i] > '9'){
            a[i] -= 10;
            carry = 1;
        }
        else carry = 0;
    }
    if (carry) a = "1" + a;
    return a;
}

string sub(string a, string b){
    if (a.size() > b.size()) b = string(a.size() - b.size(),
    '0') + b;
    if (a.size() < b.size()) a = string(b.size() - a.size(),
    '0') + a;
    long carry = 0;
    for(long i = a.size() - 1; i>=0; --i){
        a[i] -= b[i] - '0' + carry;
        if (a[i] < '0'){
            a[i] += 10;
            carry = 1;
        }
        else carry = 0;
    }
    for(long i = 0; i<a.size(); ++i) if (a[i] != '0') return
    a.substr(i);
    return "0";
}

string mul(string a, ll x){
    if (x == 0) return "0";
    string ans = "";
    ll carry = 0;
    reverse(ALL(a));
    for(auto c: a){
        long digit = c - '0';
        carry += x * digit;
        ans.push_back(carry % 10 + '0');
        carry /= 10;
    }
    while(carry > 0){
        ans.push_back(carry % 10 + '0');
        carry /= 10;
    }
    reverse(ALL(ans));
    return ans;
}

const ll BLOCK = 15;
string mul(string a, string b){
    if (a == "0" || b == "0") return "0";
    string ans = "0";
    for(long i = 0; i<(b.size() + BLOCK - 1) / BLOCK; ++i){
        long e = b.size() - i * BLOCK;
        long s = max(0, e - BLOCK);
        ll cur = 0;
        for(long j = s; j<e; ++j)
            cur = cur * 10 + (b[j] - '0');
        ans = add(ans, mul(a, cur) + string(i * BLOCK, '0'));
    }
}
```

```
return ans;
}

string fast_pow(string a, ll n){
    if (n == 0) return "1";
    string tmp = fast_pow(a, n/2);
    tmp = mul(tmp, tmp);
    if (n % 2) return mul(tmp, a);
    return tmp;
}

string di(string a, ll x){
    string ans = "";
    ll carry = 0;
    for(char c: a){
        long digit = c - '0';
        carry = carry * 10 + digit;
        ans.push_back('0' + carry / x);
        carry %= x;
    }
    for(long i = 0; i<ans.size(); ++i) if (ans[i] != '0')
    return ans.substr(i);
    return "0";
}

/*----- Big num template end
-----*/
```

7.13 Sqrt Mod

```
ll sqrt(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (modpow(a, (p - 1) / 2, p) != 1) return -1; // else no
    solution
    if (p % 4 == 3) return modpow(a, (p + 1) / 4, p);
    // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
    ll s = p - 1, n = 2;
    int r = 0, m;
    while (s % 2 == 0)
        ++r, s /= 2;
    // find a non-square mod p
    while (modpow(n, (p - 1) / 2, p) != p - 1) ++n;
    ll x = modpow(a, (s + 1) / 2, p);
    ll b = modpow(a, s, p), g = modpow(n, s, p);
    for (; r = m) {
        ll t = b;
        for (m = 0; m < r && t != 1; ++m)
            t = t * t % p;
        if (m == 0) return x;
        ll gs = modpow(g, 1LL << (r - m - 1), p);
        g = gs * gs % p;
        x = x * gs % p;
        b = b * g % p;
    }
}
```

7.14 Super Eratosthene

```
/* LOOK AT THIS DUDE! THIS SHIT IS 15 TIME FASTER THAN
ORDINARY SIEVE. */
/* IT IS THE MOST CONCISE VERSION. IT IS TOO COMPLICATED */
/*
* Usage :
* isPrime() : function checks primality using Miller-Rabin.
* calculatePhi() : function computes the Euler's Totient
Function for a range of values.
* euclid() : Computes the greatest common divisor (GCD) of
two numbers using the
extended Euclidean algorithm, also returning the coefficients
for Bézout's identity.
* update_sieve() : function marks non-prime numbers in the
current segment.
* sieve() : Uses the segmented sieve to compute all primes up
to MAX,
with wheel factorization to skip certain multiples and reduce
computation.
*/

typedef unsigned long long ull;
typedef long long ll;
```

```

const int LIM = 5000000;
const int MAX = 1000000000LL, WHEEL = 3 * 5 * 7 * 11 * 13,
SIEVE_SPAN = WHEEL * 64;
const int N_SMALL_PRIMES = 6536;
uint64_t ONES[64], si[SIEVE_SPAN / 128 + 1], pattern[WHEEL];
int phi[LIM], small_primes[N_SMALL_PRIMES];

bool isPrime(ull n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    ull A[] = {2, 325, 9375, 28178, 450775, 9780504,
1795265022},
        s = __builtin_ctzll(n - 1), d = n >> s;
    for (ull a : A) {
        ull p = modpow(a % n, d, n), i = s;
        while (p != 1 && p != n - 1 && a % n && i--) p =
            modmul(p, p, n);
        if (p != n - 1 && i != s) return 0;
    }
    return 1;
}

void calculatePhi() {
    for (int i = 0; i < LIM; ++i) phi[i] = i & 1 ? i : i / 2;
    for (int i = 3; i < LIM; i += 2) if (phi[i] == i)
        for (int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}

ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a / b * x, d;
}

inline void mark(uint64_t *s, int o) { s[o >> 6] |= ONES[o &
63]; }
inline int test(uint64_t *s, int o) { return (s[o >> 6] &
ONES[o & 63]) == 0; }

void update_sieve(int offset) {
    for (int i = 0, k; i < SIEVE_SPAN / 128 + 1; i += k) {
        k = min(WHEEL, SIEVE_SPAN / 128 + 1 - i);
        memcpy(si + i, pattern, sizeof(*pattern) * k);
    }
    if (offset == 0) si[0] |= ONES[0], si[0] &= ~(ONES[1] |
ONES[2] | ONES[3] | ONES[5] | ONES[6]);
    for (int i = 0; i < N_SMALL_PRIMES; ++i) {
        int j = small_primes[i] * small_primes[i];
        if (j > offset + SIEVE_SPAN - 1) break;
        if (j > offset) j = (j - offset) >> 1;
        else { j = small_primes[i] - offset % small_primes[i];
        if ((j & 1) == 0) j += small_primes[i]; j >>= 1; }
        while (j < SIEVE_SPAN / 2) mark(si, j), j +=
            small_primes[i];
    }
}

void sieve() {
    for (int i = 0; i < 64; ++i) ONES[i] = 1ULL << i;
    for (int i = 3; i < 256; i += 2) if (test(si, i >> 1))
        for (int j = i * i / 2; j < 32768; j += i) mark(si,
j);
    for (int i = 8, m = 0; i < 32768; ++i) if (test(si, i))
        small_primes[m++] = i * 2 + 1;
    for (int i = 1; i < WHEEL * 64; i += 3) mark(pattern, i);
    for (int i = 2; i < WHEEL * 64; i += 5) mark(pattern, i);
    for (int i = 3; i < WHEEL * 64; i += 7) mark(pattern, i);
    for (int i = 5; i < WHEEL * 64; i += 11) mark(pattern, i);
    for (int i = 6; i < WHEEL * 64; i += 13) mark(pattern, i);

    long long sum_primes = 2;
    for (int offset = 0; offset < MAX; offset += SIEVE_SPAN) {
        update_sieve(offset);
        for (uint32_t j = 0; j < SIEVE_SPAN / 128 + 1; ++j) {
            uint64_t x = ~si[j];
            while (x) {
                uint32_t p = offset + (j << 7) +
                    (__builtin_ctzll(x) << 1) + 1;
                if (p > offset + SIEVE_SPAN - 1) break;
                if (p <= MAX) sum_primes += p;
                x ^= (-x & x);
            }
        }
    }
}

```

```

    }
}
assert(sum_primes == SUM_PRIMES);
return;
}

```

8 Combinatorics

8.1 FFT (Integer)

```

/*
Usage:
- conv: multiply A and B modulo 998244353
*/

const int MOD = 998244353;

const int N = 2e5 + 69;

namespace NTT{
    const int mod = MOD;
    ll modpow(ll b, ll e) {
        ll ans = 1;
        for (; e; b = b * b % mod, e /= 2)
            if (e & 1) ans = ans * b % mod;
        return ans;
    }
    const int root = 62; // = 998244353
    // For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479
    << 21
    // and 483 << 21 (same root). The last two are > 10^9.
    typedef vector<ll> vl;

    #define rep(i, a, b) for(int i = a; i < (b); ++i)
    #define all(x) begin(x), end(x)
    #define sz(x) (int)(x).size()
    typedef long long ll;
    typedef pair<int, int> pii;
    typedef vector<int> vi;
    void ntt(vl &a) {
        int n = sz(a), L = 31 - __builtin_clz(n);
        static vl rt(2, 1);
        for (static int k = 2, s = 2; k < n; k *= 2, s++) {
            rt.resize(n);
            ll z[] = {1, modpow(root, mod >> s)};
            rep(i, k, 2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
        }
        vi rev(n);
        rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
        rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
        for (int k = 1; k < n; k *= 2)
            for (int i = 0; i < n; i += 2 * k) rep(j, 0, k) {
                ll z = rt[j + k] * a[i + j + k] % mod, &ai =
                    a[i + j];
                a[i + j + k] = ai - z + (z > ai ? mod : 0);
                ai += (ai + z >= mod ? z - mod : z);
            }
    }
    vl conv(const vl &a, const vl &b) {
        if (a.empty() || b.empty()) return {};
        int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
            n = 1 << B;
        int inv = modpow(n, mod - 2);
        vl L(a), R(b), out(n);
        L.resize(n), R.resize(n);
        ntt(L), ntt(R);
        rep(i, 0, n)
            out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv %
                mod;
        ntt(out);
        return {out.begin(), out.begin() + s};
    }
    #undef vl
}

```

8.2 FFT (Float)

```

/*
Usage:
- conv: multiply A and B modulo 998244353
*/

```

```

namespace PolyMul {

    using namespace std;
    using cd = complex<double>;
    const double PI = acos(-1);

    // Perform FFT or inverse FFT
    void fft(vector<cd>& a, bool invert) {
        int n = a.size();
        for (int i = 1, j = 0; i < n; ++i) {
            int bit = n >> 1;
            for (; j & bit; bit >>= 1) j ^= bit;
            j ^= bit;
            if (i < j) swap(a[i], a[j]);
        }

        for (int len = 2; len <= n; len <<= 1) {
            double ang = 2 * PI / len * (invert ? -1 : 1);
            cd wlen(cos(ang), sin(ang));
            for (int i = 0; i < n; i += len) {
                cd w(1);
                for (int j = 0; j < len / 2; ++j) {
                    cd u = a[i + j], v = a[i + j + len / 2] * w;
                    a[i + j] = u + v;
                    a[i + j + len / 2] = u - v;
                    w *= wlen;
                }
            }
        }

        if (invert) for (cd& x : a) x /= n;
    }

    // Multiply two polynomials
    vector<double> multiply(const vector<double>& a, const
vector<double>& b) {
        int n = 1;
        int total_size = a.size() + b.size() - 1;
        while (n < total_size) n <<= 1; // Ensure n is a
power of two

        vector<cd> fa(a.begin(), a.end()), fb(b.begin(),
b.end());
        fa.resize(n);
        fb.resize(n);

        fft(fa, false);
        fft(fb, false);
        for (int i = 0; i < n; ++i) fa[i] *= fb[i];
        fft(fa, true);

        vector<double> result(total_size);
        for (int i = 0; i < total_size; ++i) result[i] =
fa[i].real();
        return result;
    }
}

```

8.3 Matrix

```

struct Matrix {
    int n, m;
    vector<vector<Modular>> a;

    Matrix(int n, int m) : n(n), m(m), a(n,
vector<Modular>(m)) {}

    vector<Modular>& operator[] (int i) { return a[i]; }
    const vector<Modular>& operator[] (int i) const { return
a[i]; }

    void read() {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                cin >> a[i][j];
            }
        }
    }
}

```

```

void print() const {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            cout << a[i][j] << " \n"[j + 1 == m];
        }
    }
}

static Matrix eye(int n) {
    Matrix res(n, n);
    for (int i = 0; i < n; i++) {
        res[i][i] = 1;
    }
    return res;
}

Matrix operator *(const Matrix& b) {
    assert(m == b.n);
    Matrix res(n, b.m);
    for (int i = 0; i < n; i++) {
        for (int k = 0; k < b.m; k++) {
            for (int j = 0; j < m; j++) {
                res[i][k] += a[i][j] * b[j][k];
            }
        }
    }
    return res;
}

Matrix pow(long long k) const {
    assert(n == m);
    Matrix res = eye(n);
    Matrix base = *this;
    while (k > 0) {
        if (k & 1) res = res * base;
        base = base * base;
        k >>= 1;
    }
    return res;
}

Modular det() const {
    assert(n == m);
    Matrix tmp = *this;
    Modular ans = 1;

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            while (tmp[j][i].x != 0) {
                ll t = tmp[i][i].x / tmp[j][i].x;

                if (t != 0) {
                    Modular factor(t);
                    for (int k = i; k < n; k++) {
                        tmp[i][k] -= tmp[j][k] * factor;
                    }
                }

                swap(tmp[i], tmp[j]);
                ans *= Modular(-1);
            }
        }

        ans *= tmp[i][i];
        if (ans.x == 0) return 0;
    }
    return ans;
}
};

```

8.4 Modular

```

/*
Usage: - use as if it was integers
*/

const int MOD = 998244353;
struct Modular{
    ll x;
    Modular(ll _x = 0){
        x = _x % MOD;
    }
};

```

```

    if (x < 0) x += MOD;
}
Modular& operator += (Modular y){
    x += y.x;
    if (x >= MOD) x -= MOD;
    return *this;
}
Modular operator + (Modular y) {
    Modular tmp = *this;
    return tmp += y;
}

Modular& operator -= (Modular y){
    x -= y.x;
    if (x < 0) x += MOD;
    return *this;
}
Modular operator - (Modular y) {
    Modular tmp = *this;
    return tmp -= y;
}

Modular& operator *= (Modular y){
    x *= y.x;
    if (x >= MOD) x %= MOD;
    return *this;
}
Modular operator * (Modular y) {
    Modular tmp = *this;
    return tmp *= y;
}

// use at your own risk
bool operator == (Modular y){
    return x == y.x;
}
bool operator != (Modular y){
    return x != y.x;
}
};

ostream& operator << (ostream& out, Modular x){
    out << x.x;
    return out;
}

istream& operator >> (istream& in, Modular& x) {
    in >> x.x;
    return in;
}

Modular fast_pow(Modular a, ll n){
    Modular ans = 1;
    while(n > 0){
        if (n & 1) ans *= a;
        a *= a;
        n >>= 1;
    }
    return ans;
}

Modular inverse(Modular a){return fast_pow(a, MOD - 2);}

```

8.5 FFT + Poly Utilities

```

using ll = long long;
using C = complex<double>;
constexpr ll MOD = 998244353;
static inline ll modpow(ll x, ll e) {
    ll r = 1 % MOD;
    while (e) {
        if (e & 1) r = r * x % MOD;
        x = x * x % MOD;
        e >>= 1;
    }
    return r;
}

static inline ll invmod(ll x) { return modpow(x, MOD - 2); }
struct Poly {
    vector<ll> a;
    Poly(vector<ll> v = vector<ll>(0)) : a(v) {}
}

```

```

// ----- FFT (for convolution) -----

static void fft(vector<C> &f) {
    int n = (int)f.size(), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1);
    for (int k = 2; k < n; k <= 1) {
        R.resize(n); rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
        for (int i = k; i < 2 * k; ++i) rt[i] = R[i] = (i
            & 1) ? R[i/2] * x : R[i/2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; ++i) rev[i] = (rev[i/2] | (i & 1) << L) / 2;
    for (int i = 0; i < n; ++i) if (i < rev[i]) swap(f[i], f[rev[i]]);
    for (int k = 1; k < n; k <= 1)
        for (int i = 0; i < n; i += k << 1)
            for (int j = 0; j < k; ++j) {
                auto x = (double*)&rt[j+k], y =
                    (double*)&f[i+j+k];
                C z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] +
                    x[1]*y[0]);
                f[i+j+k] = f[i+j] - z;
                f[i+j] += z;
            }
}

// convolution using FFT-splitting trick (handles large MOD)
static vector<ll> conv(const vector<ll>& A, const
vector<ll>& B) {
    if (A.empty() || B.empty()) return {};
    int need = (int)A.size() + (int)B.size() - 1;
    int B2 = 32 - __builtin_clz(need), n = 1 << B2;
    int cut = (int)floor(sqrt(MOD));
    vector<C> L(n), R(n);
    for (size_t i = 0; i < A.size(); ++i) L[i] =
        C((ll)(A[i] / cut), (ll)(A[i] % cut));
    for (size_t i = 0; i < B.size(); ++i) R[i] =
        C((ll)(B[i] / cut), (ll)(B[i] % cut));
    fft(L); fft(R);
    vector<C> outl(n), outs(n);
    for (int i = 0; i < n; ++i) {
        int j = (-i) & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) /
            C(0, 1);
    }
    fft(outl); fft(outs);
    vector<ll> res(need);
    for (int i = 0; i < need; ++i) {
        ll av = (ll) (real(outl[i]) + 0.5) % MOD;
        ll cv = (ll) (imag(outs[i]) + 0.5) % MOD;
        ll bv = ((ll)(imag(outl[i]) + 0.5) +
            (ll)(real(outs[i]) + 0.5)) % MOD;
        res[i] = ((av * cut + bv) % MOD) * cut + cv) %
            MOD;
    }
    return res;
}

// ----- operators -----
Poly operator+(const Poly& o) const {
    vector<ll> r(max(a.size(), o.a.size()), 0);
    for (size_t i = 0; i < r.size(); ++i) {
        ll x = i < a.size() ? a[i] : 0;
        ll y = i < o.a.size() ? o.a[i] : 0;
        r[i] = (x + y) % MOD;
    }
    return Poly(r);
}

Poly operator-(const Poly& o) const {
    vector<ll> r(max(a.size(), o.a.size()), 0);
    for (size_t i = 0; i < r.size(); ++i) {
        ll x = i < a.size() ? a[i] : 0;
        ll y = i < o.a.size() ? o.a[i] : 0;
        r[i] = (x - y) % MOD;
        if (r[i] < 0) r[i] += MOD;
    }
}

```

```

    return Poly(r);
}
Poly operator*(const Poly& o) const {
    return Poly(conv(a, o.a));
}

// ----- basic utilities -----
int deg() const { return (int)a.size() - 1; }
void mod_xk(int k) { if ((int)a.size() > k) a.resize(k); }

// ----- calculus -----
Poly deri() const {
    if (a.size() <= 1) return Poly();
    vector<ll> r(a.size()-1);
    for (size_t i = 1; i < a.size(); ++i) r[i-1] = a[i] *
        (1l)i % MOD;
    return Poly(r);
}
Poly inte() const {
    vector<ll> r(a.size()+1);
    r[0] = 0;
    for (size_t i = 0; i < a.size(); ++i) r[i+1] = a[i] *
        invmod(i+1) % MOD;
    return Poly(r);
}

// ----- inverse, log, exp -----
// inverse assumes a[0] != 0
Poly inv() const {
    assert(!a.empty() && a[0] != 0);
    Poly R(vector<ll>{invmod(a[0])});

    int n = (int)a.size();
    int d = 1;
    while (d < n) {
        d <= 1;
        vector<ll> f(d);
        for (int i = 0; i < (int)min(f.size(), a.size());
            ++i) f[i] = a[i];
        Poly F(f);

        Poly FR = F * R;
        FR.mod_xk(d);
        for (auto &x : FR.a)
            x = (MOD - x) % MOD;
        FR.a[0] = (FR.a[0] + 2) % MOD;
        R = R * FR;
        R.mod_xk(d);
    }
    R.mod_xk(n);
    return R;
}

// log assumes a[0] == 1
Poly log(int k = -1) const {
    int need = (k == -1 ? (int)a.size() : k);
    assert(!a.empty() && a[0] == 1);
    Poly res = (deri() * inv()).mod_xk_here(need).inte();
    res.mod_xk(need);
    return res;
}

// helper for chaining mod_xk without mutating original
Poly mod_xk_here(int k) const { Poly r = *this;
    r.mod_xk(k); return r; }

Poly exp() const {
    assert(!a.empty() && a[0] == 0);
    Poly R({1});
    int n = a.size();
    for (int len = 2; ; len <= 1) {
        R.a.resize(len, 0);
        Poly Ln = R.log(len);
        Ln.a[0] = 1;
        for (int i = 1; i < len; ++i) {
            ll val = (i < n ? a[i] : 0);
            Ln.a[i] = val - Ln.a[i];
            if (Ln.a[i] < 0) Ln.a[i] += MOD;
        }
        R = R * Ln;
        R.mod_xk(len);
    }
}

```

```

        if (len >= n) break;
    }

    R.mod_xk(n);
    return R;
}

};

8.6 Xor Convolution

/*
Xor convolution:
- Usage: xormul(a, b, &c);
*/

const int mod = 998244353;

int inverse(int x, int mod) {
    return x == 1 ? 1 : mod - mod / x * inverse(mod % x, mod)
        % mod;
}

void xormul(vector<int> a, vector<int> b, vector<int> &c) {
    int m = (int) a.size();
    c.resize(m);
    for (int n = m / 2; n > 0; n /= 2)
        for (int i = 0; i < m; i += 2 * n)
            for (int j = 0; j < n; j++) {
                int x = a[i + j], y = a[i + j + n];
                a[i + j] = (x + y) % mod;
                a[i + j + n] = (x - y + mod) % mod;
            }
        for (int n = m / 2; n > 0; n /= 2)
            for (int i = 0; i < m; i += 2 * n)
                for (int j = 0; j < n; j++) {
                    int x = b[i + j], y = b[i + j + n];
                    b[i + j] = (x + y) % mod;
                    b[i + j + n] = (x - y + mod) % mod;
                }
            for (int i = 0; i < m; i++)
                c[i] = a[i] * b[i] % mod;
        for (int n = 1; n < m; n *= 2)
            for (int i = 0; i < m; i += 2 * n)
                for (int j = 0; j < n; j++) {
                    int x = c[i + j], y = c[i + j + n];
                    c[i + j] = (x + y) % mod;
                    c[i + j + n] = (x - y + mod) % mod;
                }
    int mrev = inverse(m, mod);
    for (int i = 0; i < m; i++)
        c[i] = c[i] * mrev % mod;
}

```